CHAPTER

ATA 61

PROPELLER

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Propellers - General

Introduction

The propeller is a six bladed, constant speed, variable pitch propeller which can be feathered and used in reverse pitch. The propeller is installed on the gearbox/propeller driveshaft flange and turns clockwise when seen from the rear.

An overspeed governor, a pitch control unit and a feathering pump supply synthetic lubricating oil to operate the pitch control mechanism. The beta tubes supply the synthetic lubricating oil from the pitch control unit to the propeller operating cylinder and piston.

General Description

Refer Figure 61-1

The Dowty Aerospace Propellers CR408/6−123−F/17 propeller converts the PW150A engine torque into thrust for aircraft propulsion in flight and during ground maneuvers.

The propeller can be described by its model number as follows:

- C – Civil
- R – Dowty Aerospace Propellers
- 408 – Aircraft type identification
- 6 – Number of blades
- 123 – Blade root end size in mm.
- F – Flange mounted
- 17 – Function / Installation characteristics

The propeller has 6 all composite blades and is 13.5 ft. (4.11 m) in diameter. The blade pitch is varied from full feather to full reverse by an electronically controlled hydraulic pitch actuation system.

Pitch change is accomplished by metered oil pressure. The oil pressure originates from the related engine and is supplied to a pump which is part of the Overspeed Governor (OSG) installation. The Overspeed Governor and Pump are mounted on, and driven by, the reduction gearbox of each engine.

The propeller blades are counter weighted to move towards coarse pitch in the event of oil pressure failure.

The major components of the propeller are:

- Blades
- Hub
- Crosshead
- Beta Tube Assembly
- Pitch Change Cylinder and Piston
- Blade Bearings
- Blade Counterweights

The spinner provides an aerodynamic fairing that covers the hub assembly.

The propeller system has the following subsystems:

- Propeller
- Propeller Controlling

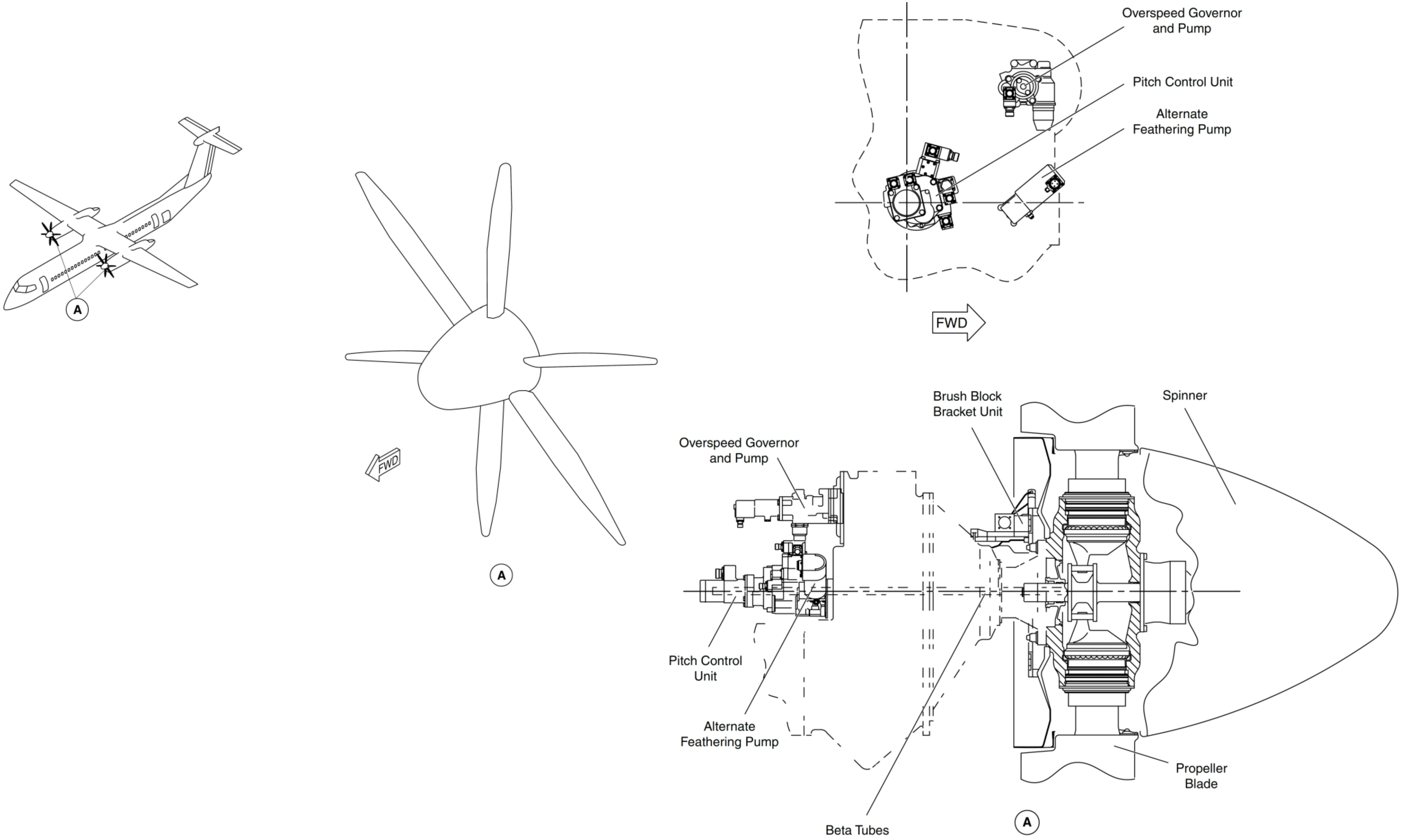


Figure 61-1 Propeller Assembly

Detailed Description

Refer Figure 61-2

The propeller is flange mounted on the front of the engine reduction gearbox, aligned by three dowels and a central mounting spigot on the propeller shaft.

Torque is transmitted from the reduction gearbox, approximately 50% through the dowels and 50% through friction of the flange interface.

A composite shim is interposed between the RGB and the propeller flange. The dowel holes are lined with composite material to limit fretting.

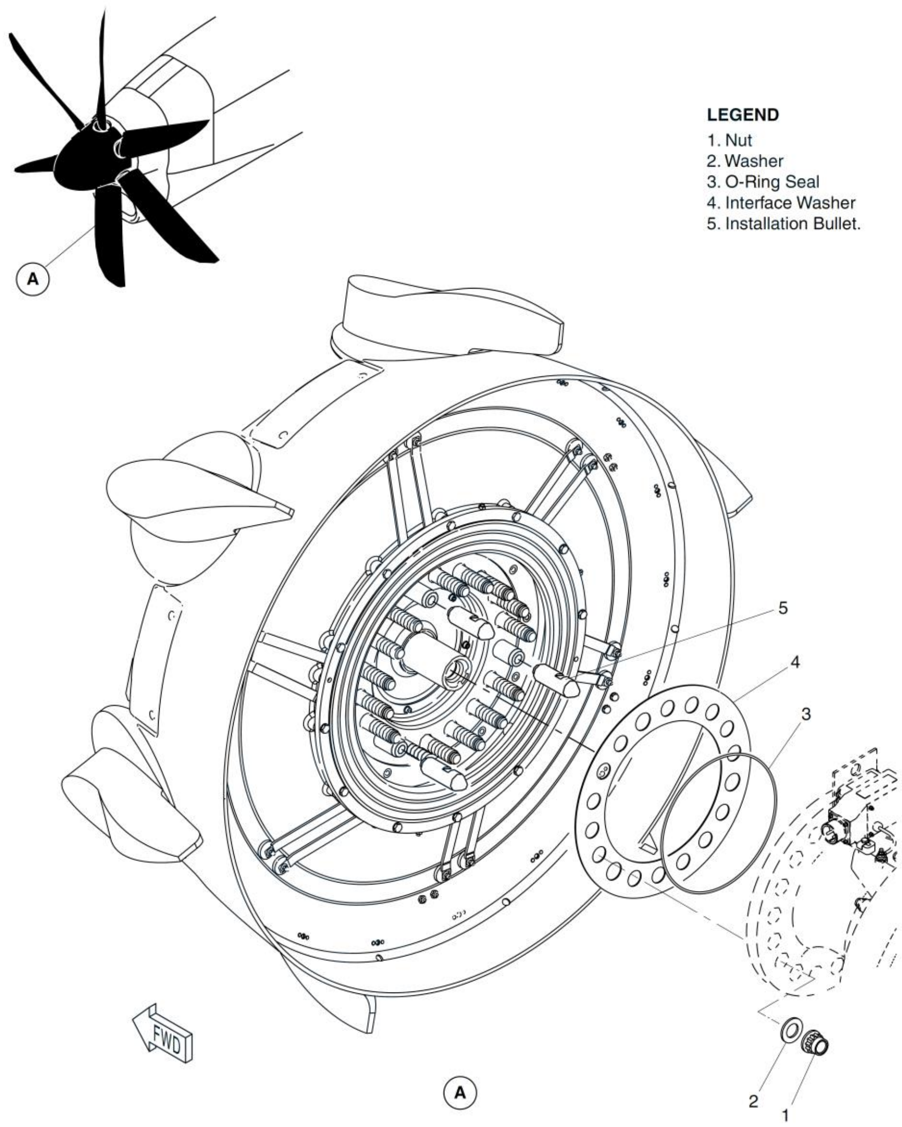


Figure 61-2 Propeller Mounting Locator

Propeller Electronic Control Unit

Refer Figures 61-3

The Propeller Electronic Control (PEC) is mounted on the spine above the engine between the center and front engine support frames.

The PEC is electronically connected to the following components:

− Condition lever
− Power lever (through FADEC)
− Magnetic Pick−up Unit (MPU),
− Beta Feedback Transducer (BFT)
− Pitch Control Unit (PCU)
− Full Authority Digital Electronic Control (FADEC)

The Propeller Electronic Control (PEC) is a dual channel microprocessor−based controller which uses inputs from the aircraft, propeller control system sensors, and the engine control system to control propeller pitch and speed. The PECs for both propeller systems are mounted in their respective engine nacelles.

Each unit has an independent circuit that performs a number of safety functions including Autofeather and Automatic Underspeed Propeller Control (AUPC). It also provides an UPTRIM command to the Automatic Take−Off Power Control System (ATPCS) of the remote engine. All of these functions are isolated from the basic control functions of the PEC.

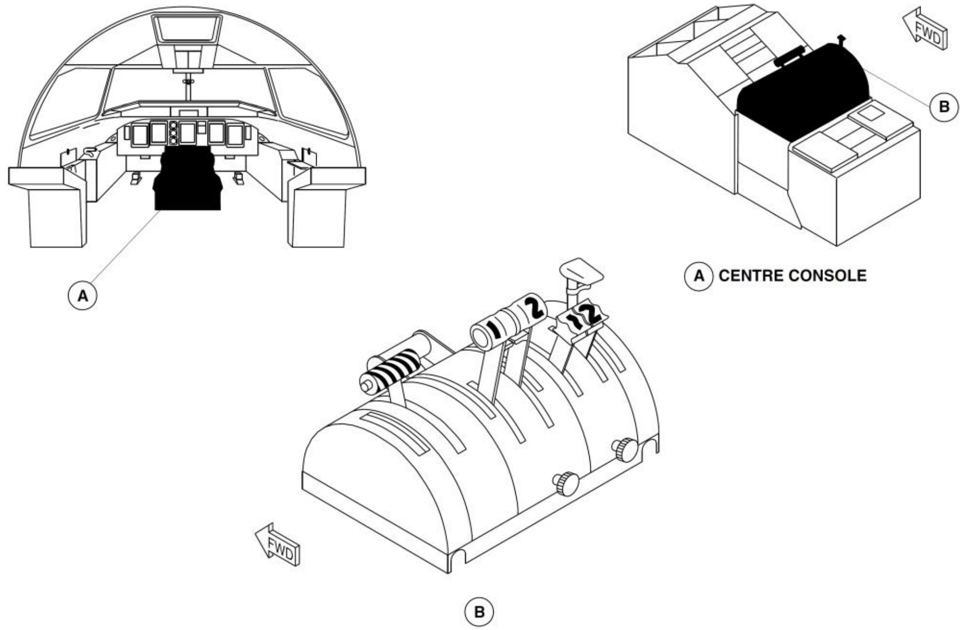


Figure 61-3 Condition Lever and Power Levers

Pitch Control Unit

Refer Figure 61-4

The Pitch Control Unit (PCU) is mounted on the rear face of the reduction gearbox, and is connected to the propeller by a dual concentric beta tube assembly.

The propeller Pitch Control Unit (PCU) is a hydromechanical device that interfaces with the propeller. Commanded electrically by the PEC, the PCU meters high pressure engine oil to a two stage servo valve mounted on the PCU and controls the flow of high pressure oil into the fine or coarse pitch chambers of the propeller pitch change cylinder as directed by the PEC so that the blades move in the desired direction. The PCU provides:

- Governed Constant Speed Operation
- Power Lever Controlled Beta Range (Flight Idle to Reverse)
- Manual Feather
- Unfeather
- Propeller Synchrophasing

High Pressure PCU Pump and O/S Governor

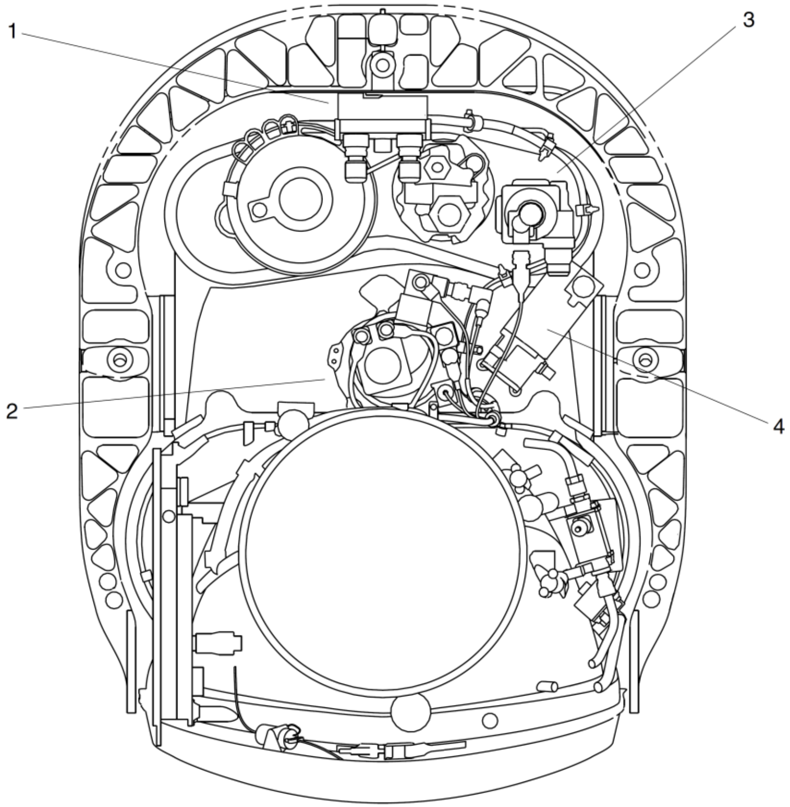
The High Pressure PCU Pump/Propeller Overspeed Governor Unit provides the PCU with high pressure oil from the engine gearbox. The High Pressure pump is a fixed displacement gear pump, driven from the reduction gearbox. The Propeller Overspeed Governor Unit is an independent mechanical system used to protect the propeller from overspeeding. The O/S Governor is a flyweight design, driven directly from the driver gear of the pump.

Auxiliary Feathering Pump

The Auxiliary Feathering Pump Unit provides an independent means of feathering the propeller in the event of a failure of the primary means of feather. The auxiliary pump consists of an electrical motor driving an external gear pump which supplies a secondary source of pressurized oil for feathering the propeller.

Propeller Modes

The PCU provides for governed constant speed operation through a propeller governor controlled by the condition levers. The power levers control blade angle in the beta range. The manual feather mode is controlled by the condition levers or by the autofeather/alternate feather system.



LEGEND

1. Propeller Electronic Control.
2. Propeller Control Unit.
3. Overspeed Governor and PCU Pump.
4. Auxiliary Feather Pump.

Figure 61-4 PCU, O/S Governor & Feathering Pump Locations

Propeller Assembly

Introduction

The propeller assembly converts engine supplied power into usable thrust for aircraft propulsion during flight and ground maneuvers.

General Description

The Pitch Control Unit supplies oil to control propeller pitch, in response to signals from the Propeller Electronic Controller (PEC). Blade pitch change is controlled by the PCU scheduling high pressure oil to the fine or coarse side of the pitch change piston through beta tube assemblies.

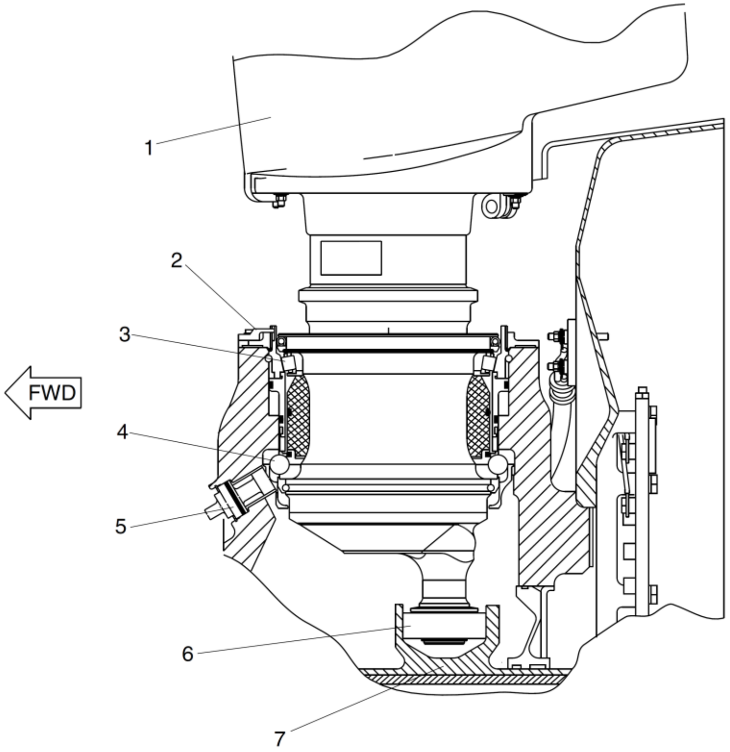
The propeller assembly has the components that follow:

- Propeller Blade Assembly
- Hub Assembly
- Spinner Assembly

Detailed Description

Refer Figure 61-5

The six bladed propeller is constant speed, variable pitch, with full feathering and full reverse capabilities. The propeller is installed on the engine reduction gearbox propeller drive flange by 15 threaded studs and nuts. The three location dowels help with the correct propeller installation and transmit some of the drive torque. The blades are installed in a flange mounted aluminum hub. Installed on the hub is a cylinder assembly which houses the pitch change actuator. A spinner is attached to the propeller composite backplate.



LEGEND

1. Blade.
2. Lockpiece.
3. Outer (Roller) Bearings.
4. Inner (Ball) Bearings
5. Plug.
6. Track Roller.
7. Crosshead.

Figure 61-5 Propeller Blade Installation Location

Propeller Blade Assembly

Refer Figures 61-6

Each blade is an all composite aerofoil construction with a steel outer root sleeve. The aerofoil has a foam core and twin carbon fibre spars with an overall braided carbon/glass fibre shell. A polyurethane spray coat for erosion protection is applied to the complete blade surface. A nickel leading edge guard is installed for blade erosion protection. A heater element to de−ice 70% of the blade radius is installed on the blade leading edge. A front−mounted aluminum alloy piston and cylinder and steel crosshead/shaft change blade pitch. The operating pins on the base of each blade engage between the faces of the crosshead, to give blade rotation.

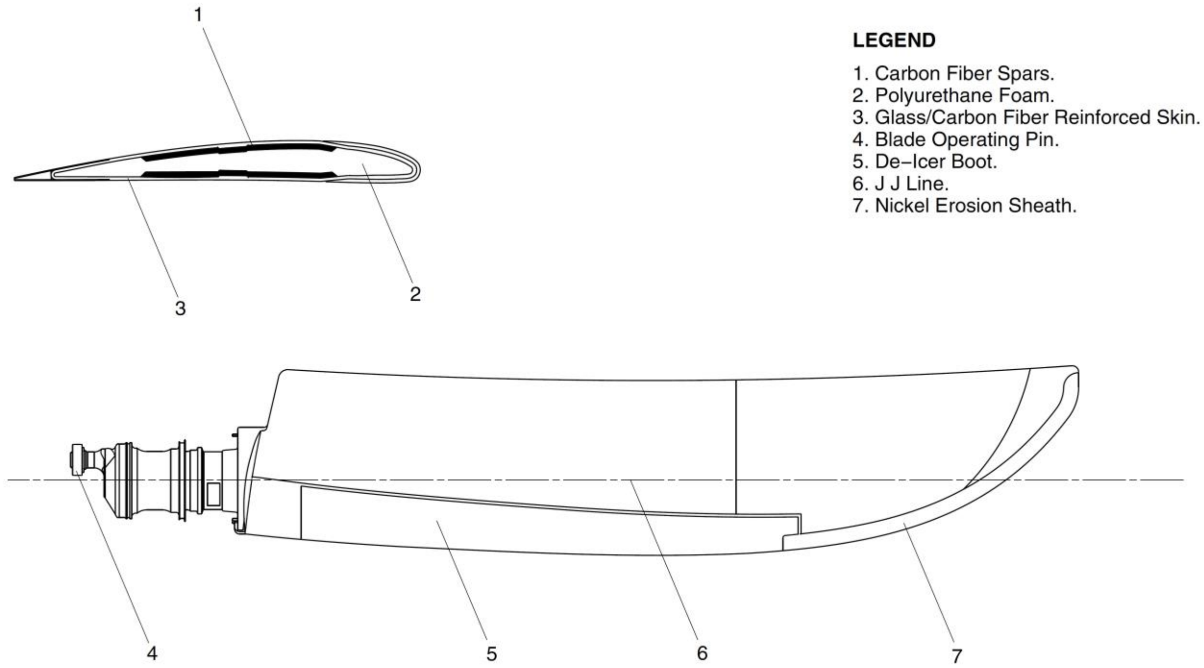


Figure 61-6 Propeller Blade Assembly Details

Actuator and Backplate Hub Assembly

Actuator

Refer Figures *61-7*

The pitch change actuator is housed in a cylinder assembly secured to the hub.

The cylinder is made of aluminum alloy and has a removable cap at the front end.

The piston is inside the cylinder forward of the crosshead assembly and held in position by a key and keyway. The piston is fitted with a "LEE" jet which allows a small continuous flow of hot oil to flow from one side to the other to keep the assembly warm in flight. The actuator controls the pitch of the blade.

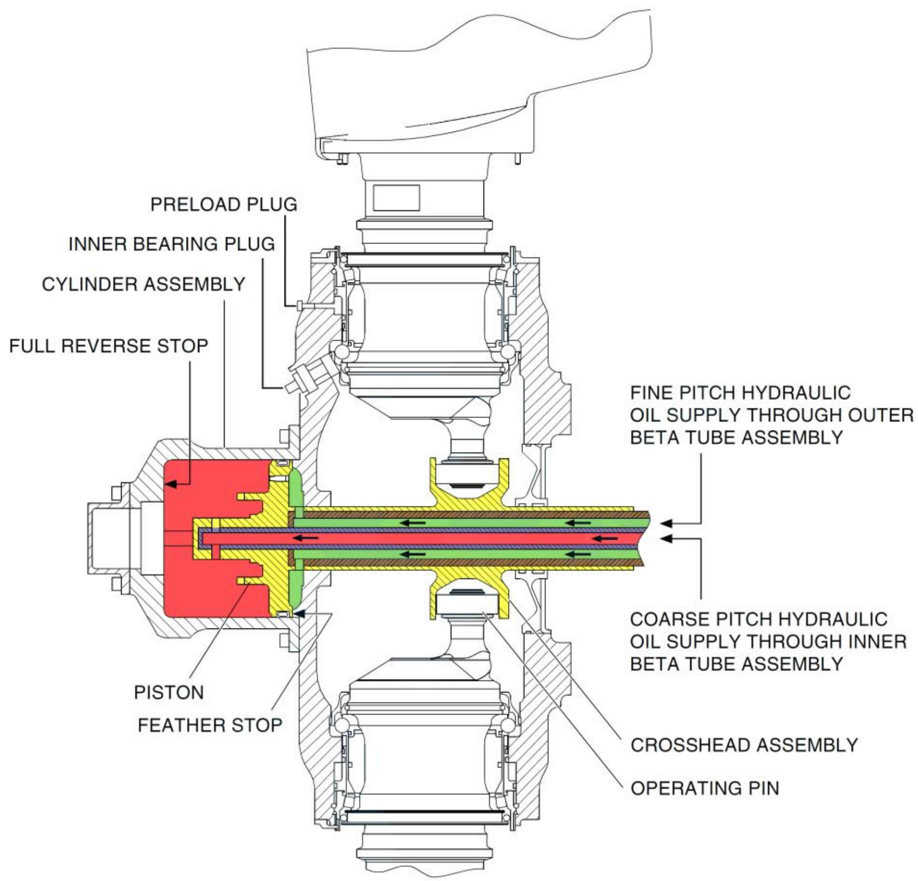


Figure 61-7 Pitch Change Actuator

Hub Assembly

Refer Figures 61-8 & *61-9*

The hub assembly is mounted on the Propeller flange of the RGB.

The hub is manufactured from a single piece aluminum alloy. The forward face has 12 equally spaced holes for attaching the cylinder. The aft face of the hub has 15 integral steel mounting studs and three location/drive dowels. The nuts on the studs have a minimum running torque of 50 lbs/in.

The hub supports six blades and has six pairs of blade root bearings races. A front-mounted aluminum alloy piston and cylinder and steel crosshead/shaft change blade pitch.

To prevent fretting at the shaft to the propeller interface, a composite shim is installed on the flange.

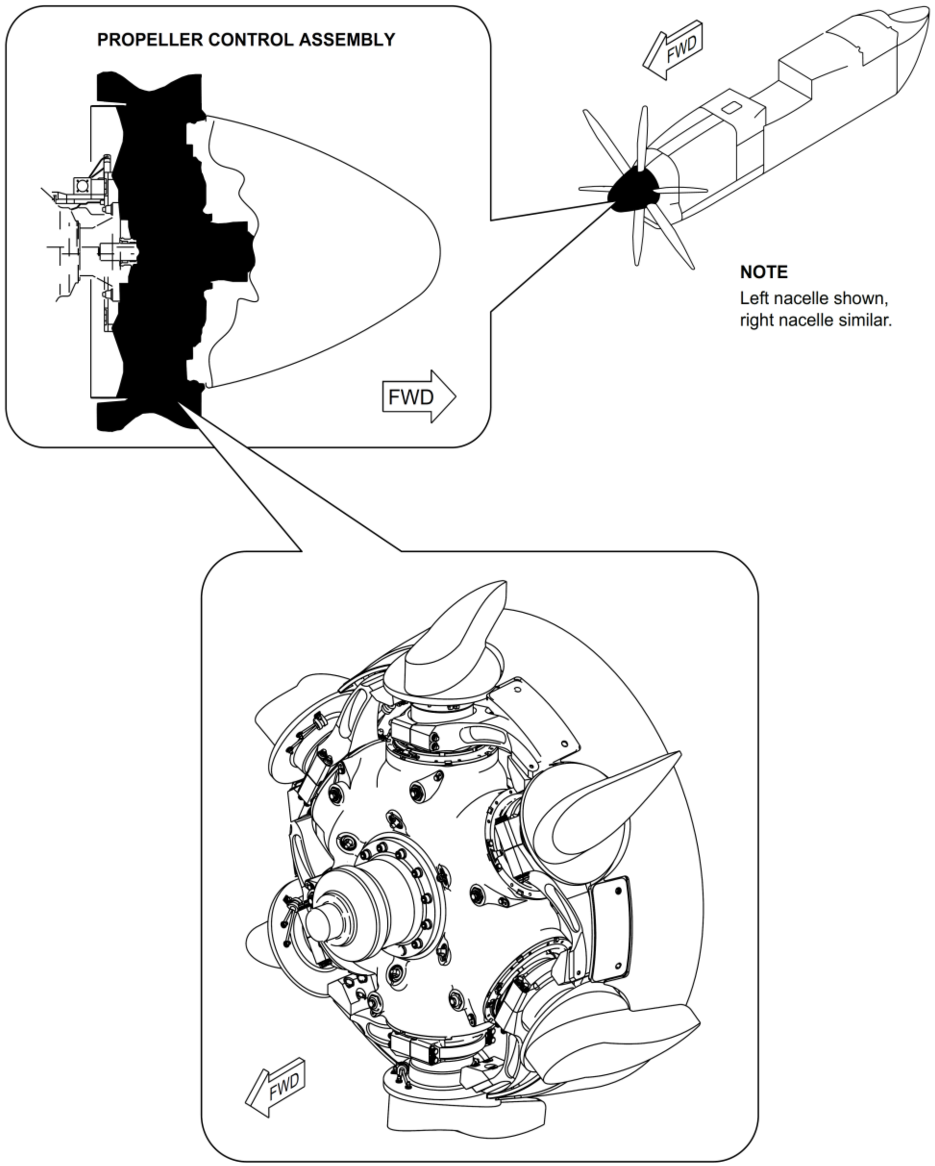


Figure 61-8 Propeller Hub Assembly

Backplate Assembly

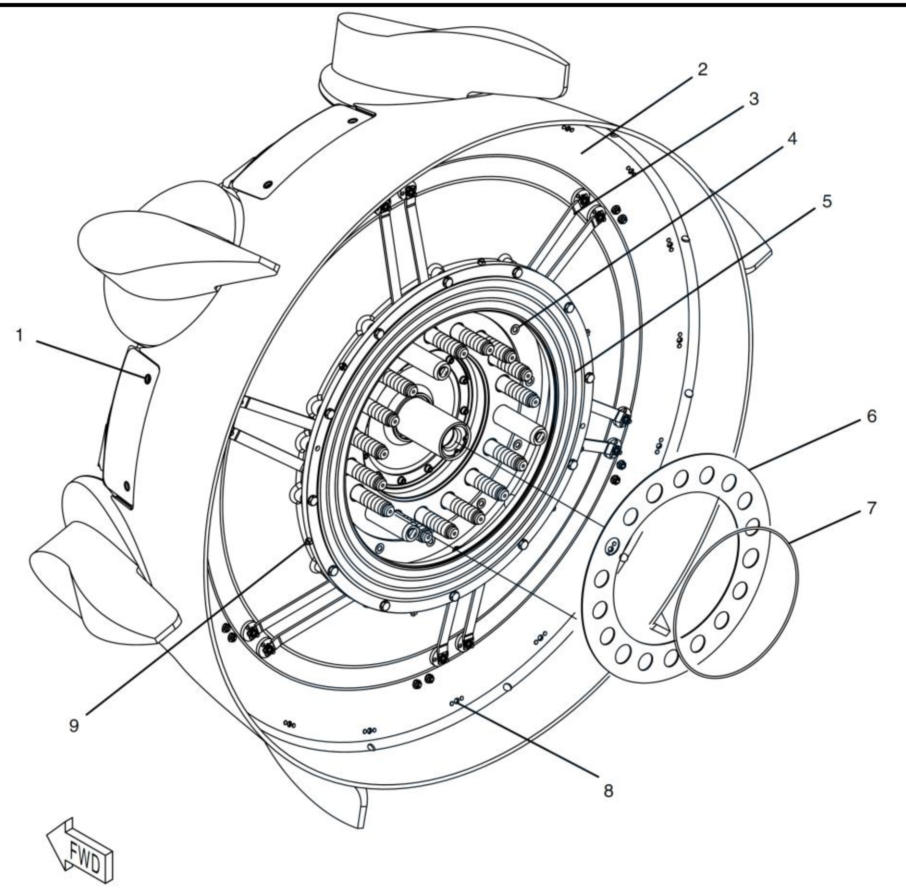
Refer Figure 61-9

The backplate assembly is installed on the rear of the hub assembly.

The backplate is constructed of carbon fiber composite. The slip ring is installed directly onto the backplate and has an aluminum alloy housing with three bronze rings in a plastic molding. The target screws that supply propeller speed and phase angle feedback are located on the external diameter of the slipring.

The backplate forms the aerodynamic interface between the spinner and engine nacelle.

The slipring is used to transfer electrical power for blade de-icing.



LEGEND
1. Spinner retention nut location (typical)
2. Backplate
3. Deice bus bars (typical)
4. Backplate attachment bolt (typical)
5. Deice slip ring
6. Gasket
7. O-ring seal
8. Static/dynamic balance weight location (typical)
9. MPU target screw (typical)

Figure 61-9 Propeller Attachment and Backplate Assembly

Spinner Assembly

Refer Figure 61-10

The spinner is installed on the propeller back-plate assembly with 12 quick release fasteners.

It is constructed of composite material made in three pieces.

The spinner gives an aerodynamic fairing over the front end of the propeller.

A centralizing/support diaphragm on the pitch change cylinder makes sure that the spinner runs correctly, when rotating with the propeller.

It is balanced by the addition of rubber pads stuck to the inner surface. Although it is a separately balanced component, it is marked by the manufacturer with a blade #1 position.

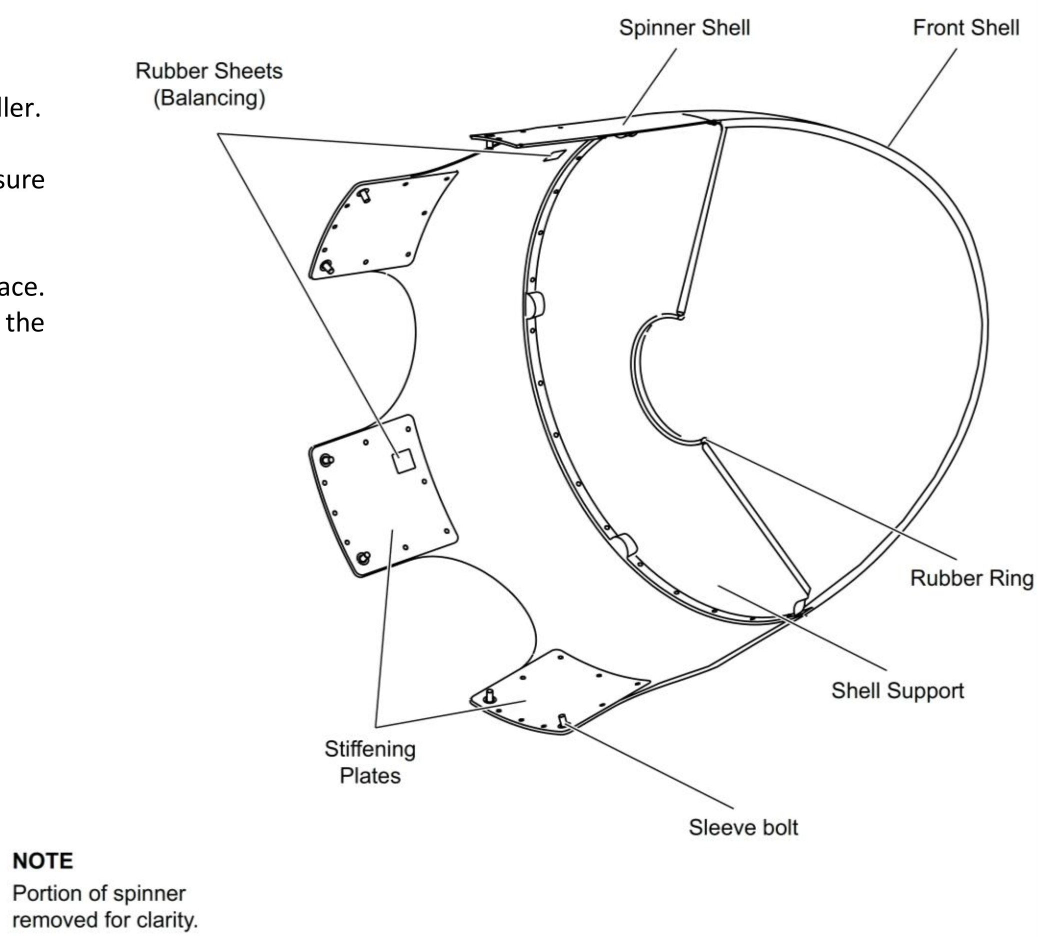


Figure 61-10 Propeller Spinner

Propeller Controlling System

Introduction

The
propeller control system modulates blade angle or pitch, to achieve the necessary propeller RPM (Np) in flight and control pitch/thrust on the ground. The propeller control system also feathers the propeller, when the engine is shut down.

General Description

Refer Figure 61-11 & 61-12

The power levers are used to control the propeller pitch, the condition levers are used to set Np and to feather the propellers. The Propeller Electronic Controller (PEC) supplies the current to the servo valve drive, through a dual line hydromechanical actuation system, to change the blade pitch.

The main modes of propeller operation for control during flight and on the ground are:

− Constant Speed Mode (Power Lever between Flight Idle and MAX Power)
− Beta Control Mode (Power Lever below Flight Idle to Max Reverse)
− Reverse speed control mode
− Synchrophase control mode.

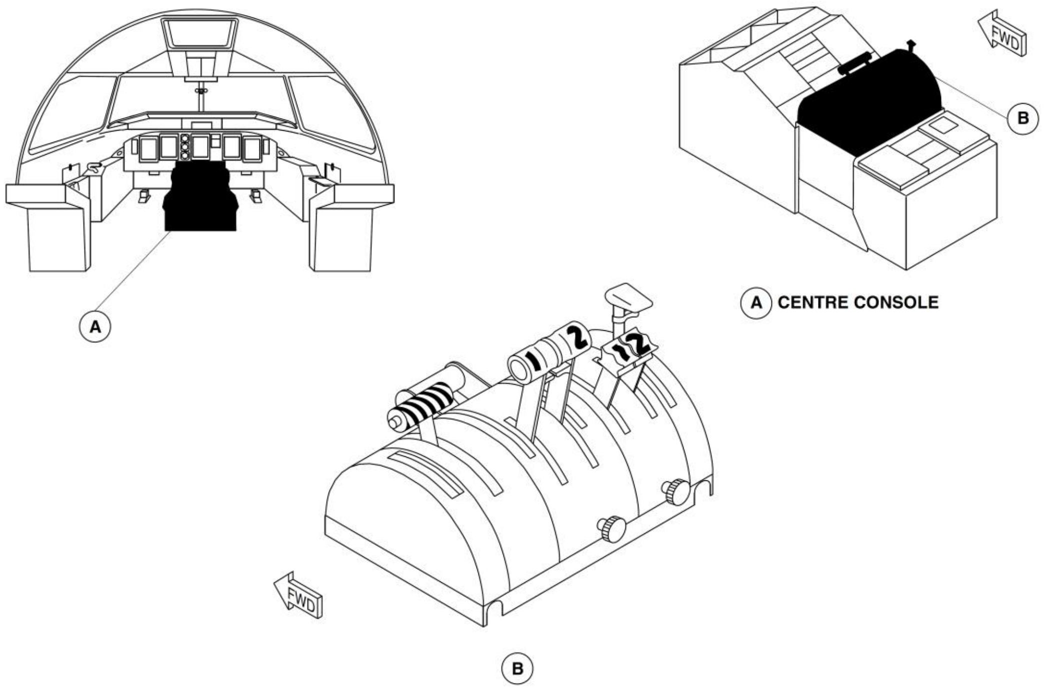


Figure 61-11 Power Quadrant

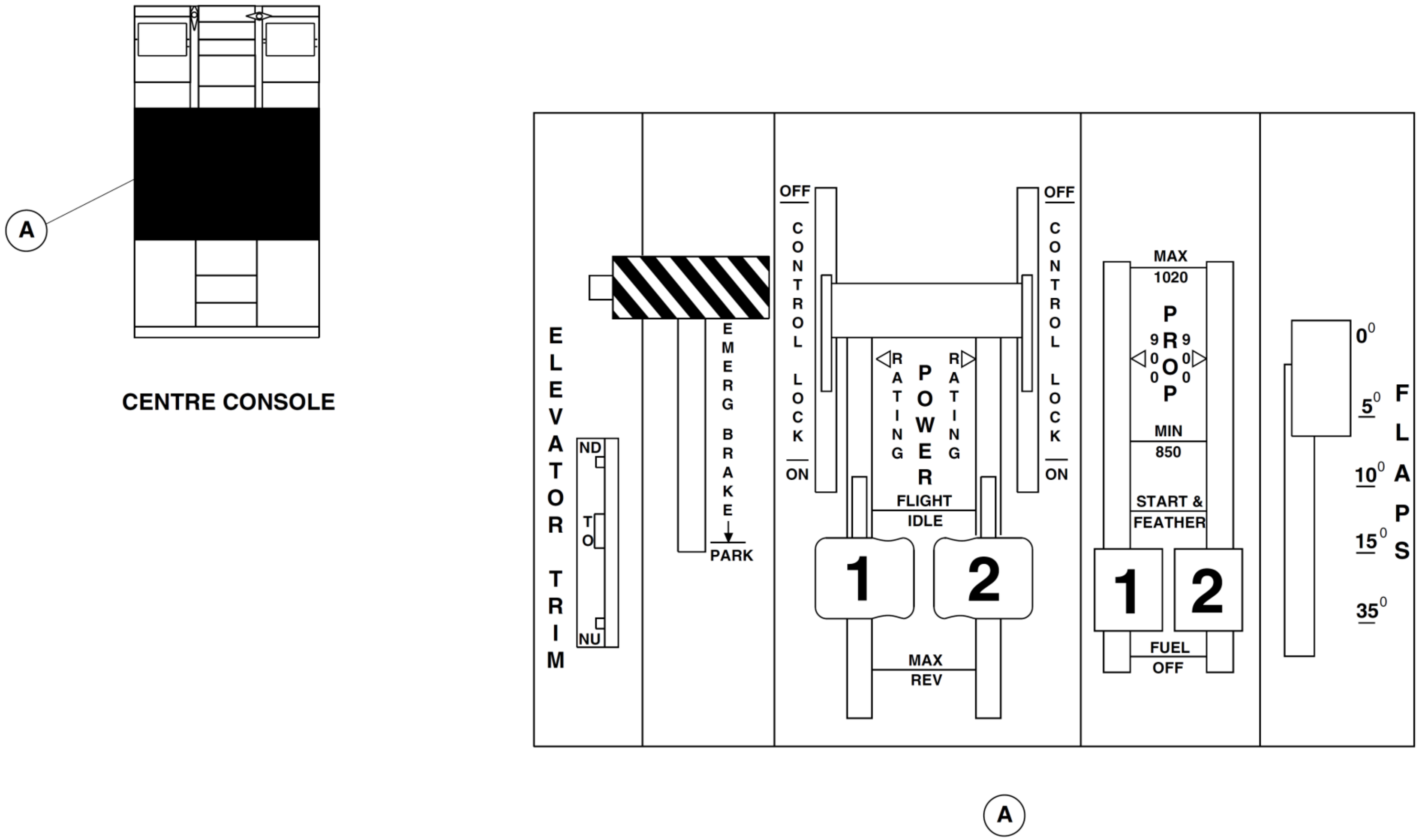
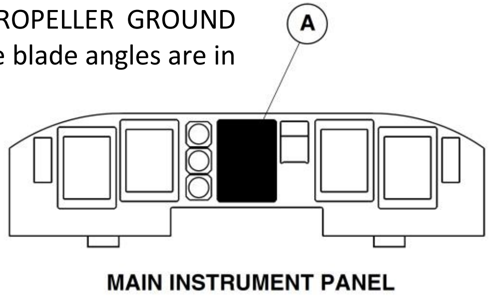
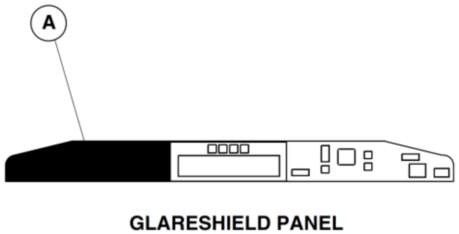


Figure 61-12 Power Quadrant, Power Levers and Condition Levers

Refer Figure 61-13 & 61-14

Propeller RPM is shown on the Engine Display (ED). PROPELLER GROUND RANGE lights on the left glareshield panel show when the blade angles are in the ground operating range.





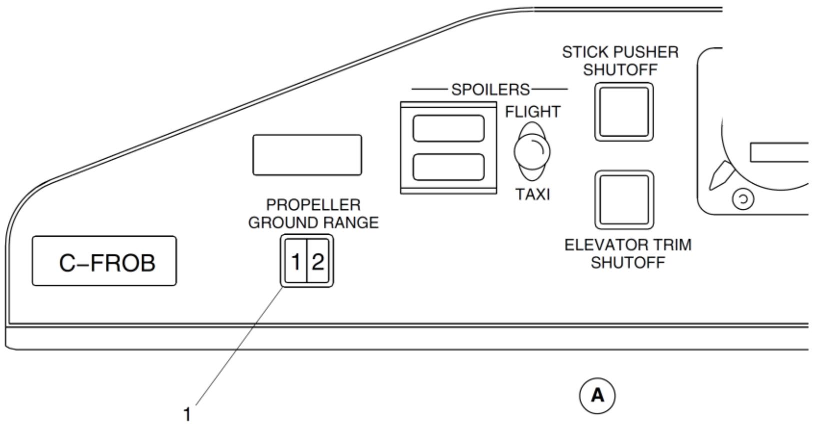
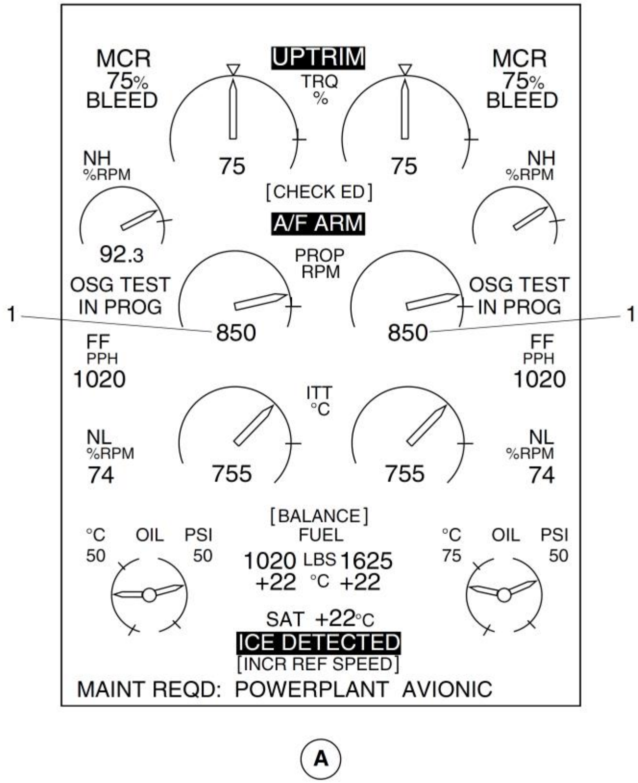


Figure 61-13 Propeller Ground Range Lights



LEGEND

1. Left/Right Propeller Speed Indication.

Figure 61-14 Propeller RPM Indications

Detailed Description

The main modes of propeller operation for control during flight and on the ground are:

- Constant Speed Mode (PLA between above Flight Idle and Rated Power Detent)
- Beta Control Mode (PLA between above Flight Idle to below Disc)
- Reverse speed control mode (PLA between below Disc and Max Reverse)
- Synchro-phase control mode (slave propeller)
- Feather Mode

Refer Figure 61-15

The propeller control system has the components that follow:

− Beta Tube Assembly
− Magnetic Pick−up Unit
− Pitch Control Unit
− Ground Beta Enable Solenoid
− Unfeather Solenoid
− Overspeed Governor— High Pressure Pump
− Feathering Pump
− Propeller Electronic Control Unit
− Propeller Control Panel

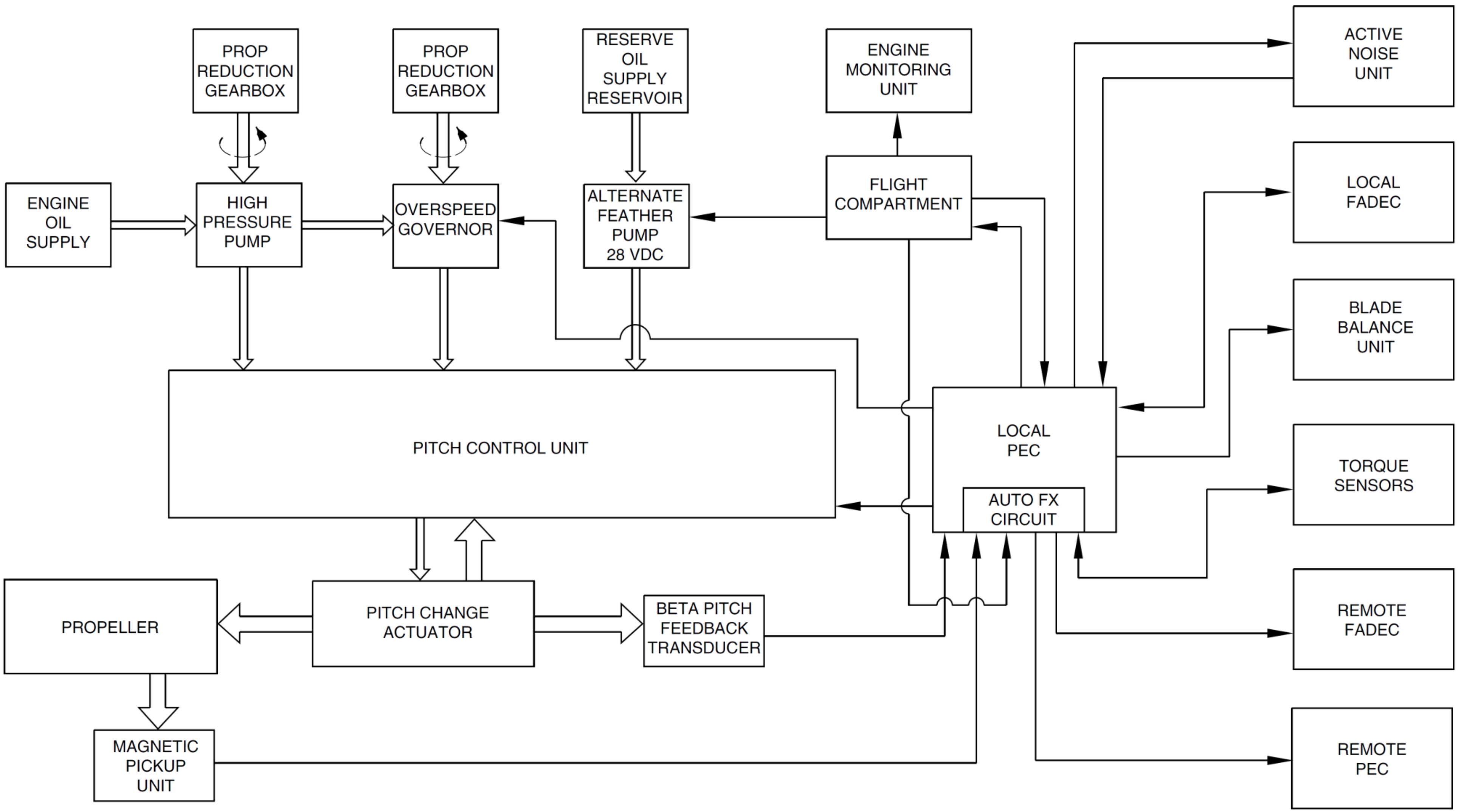


Figure 61-15 Propeller Control Block Diagram

Constant Speed Mode

Refer Figure 61-16

During in−flight "constant speeding" operation, the Propeller Electronic Control Unit (PEC) directs the servo valve to meter high pressure oil into the propeller fine pitch chamber. This is to balance the coarse−seeking moment applied to the blades, so that the propeller stays at the selected speed (Np). If there is a loss of high pressure oil supply, the blades will "autocoarsen" to a safe high pitch underspeeding condition, to give low windmilling drag.

For underspeeding conditions, other than loss of oil pressure or servo valve failure, high pressure oil is sent to the fine pitch chamber to restore propeller speed. If the Np is greater that the demanded speed, the servo valve will send the oil pressure to the coarse pitch chamber to reduce propeller speed.

Constant speed mode is entered when propeller speed reaches 850, 900 or 1020 rpm, depending on the condition lever selection.

High pressure oil for constant speeding flows through the Overspeed Governor (OSG) before it reaches the servo valve. A spool in the OSG is held at either end by the opposing forces of a spring and the toes of a pair of flyweights. The flyweights are held in a carrier which is driven round with the high pressure pump by the Propeller Reduction Gearbox (PRG).

If the servo valve sticks at the fine pitch selection, Np will increase to approximately 104%. The OSG spool will then isolate the propeller control system from the high pressure oil supply. Np will decrease due to propeller counterweight action. The OSG will then reconnect the high pressure oil supply and a stable governing condition at 104% Np will be quickly achieved.

The OSG spool spring is located in a cylinder, on a piston, that is connected to the high pressure oil supply, through a solenoid operated pilot valve. The solenoid is energized by the PROP O/SPEED GOVERNOR TEST switch on the pilot’s side console. A Weight−On−Wheels (WOW) input to the PEC prevents test operation during flight.

The Propeller Electronic Control (PEC) is a dual lane control device, that supplies the servo valve drive current, to drive two motor coils in the servo valve. A number of failure conditions will cause the propeller to go into coarse pitch automatically. The coarse pitch bias was chosen because, total loss of electrical power will cause fuel supply to the engine to be shut off, so it is appropriate in this condition to feather the propeller.

The failure conditions are as follows:

− Total loss of electrical power
− Loss of output from both PEC control lanes
− Failure of both servo valve torque motor coils

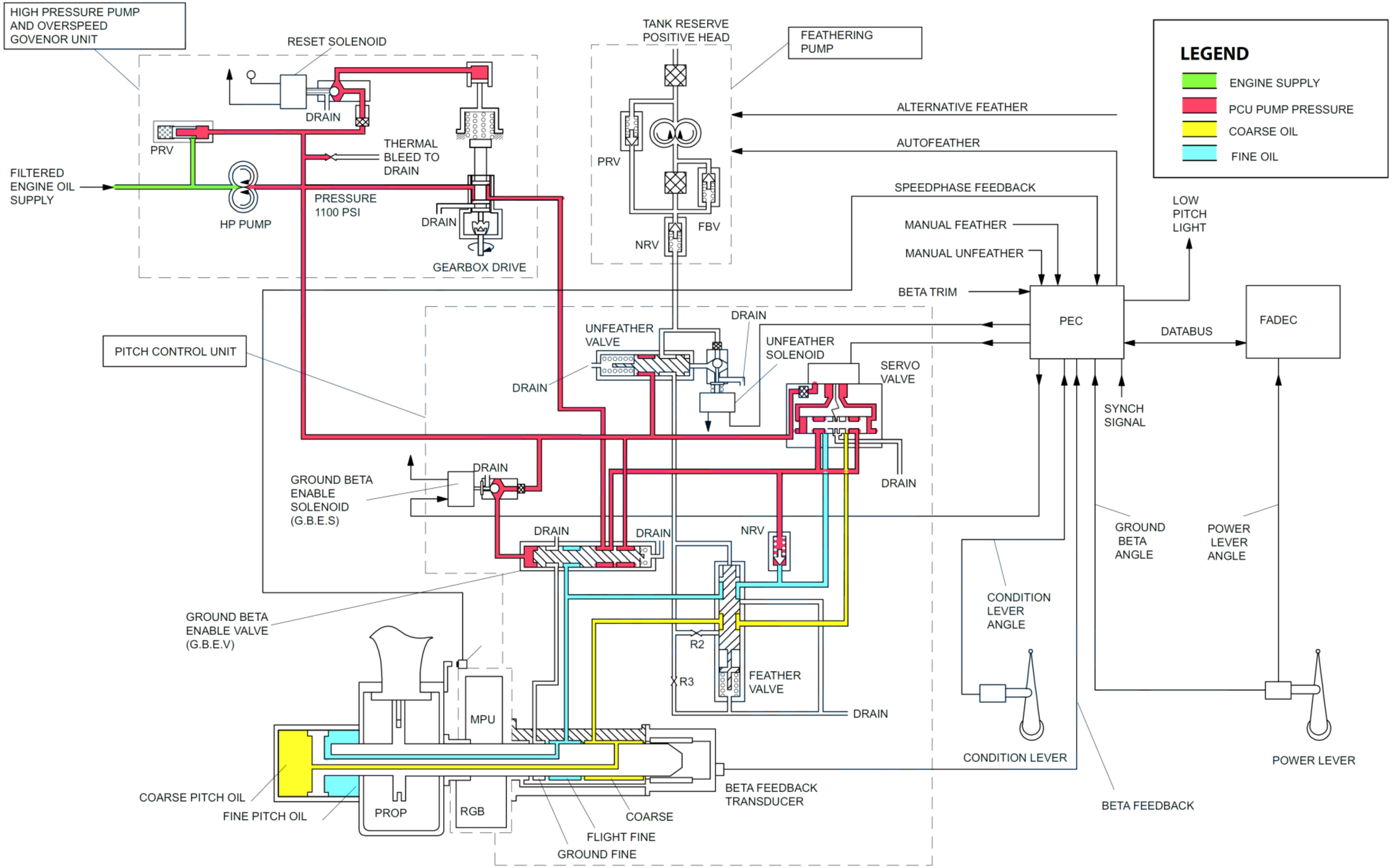


Figure 61-16 Steady State Constant Speed Control

Beta Control

Refer Figure 61-17 & 61-18

During on−ground "Beta Control" (power lever below flight idle), the Propeller Electronic Control (PEC) directs the servo valve to meter oil into the fine or coarse pitch chamber to achieve the required blade angle. The blade angle is set by the dual Power Lever Angle (PLA) Rotary Variable Differential Transformer (RVDT), located on the power lever quadrant. The PEC receives PLA signals through the Full Authority Digital Electronic Control (FADEC).

NOTE

During approach, when airspeed is relatively low, it is possible for the propeller to enter beta control. This occurs when the blade angle drops below the pitch PLA schedule and propeller RPM equals the set RPM. As soon as propeller RPM increases, the PEC automatically re−enters the constant speed mode.

The position of the ports in the PCU/Beta tubes make sure that fine pitch in the in−flight constant speed mode is limited to a blade angle of 16 degrees. This hydraulic cut−off of oil pressure is specified as the "hydraulic flight fine stop" interlock.

The flight fine stop keeps a minimum pitch consistent with positive counterweight effort driving towards coarse pitch, to make sure the OSG is effective throughout the in−flight pitch range.

A "soft" flight fine stop at approximately 16.5 degrees is programmed into the PEC. This stop makes sure the blade angle does not drop below 16.5 degrees while the power lever is at, or above flight idle in flight. This is specified as a "software flight fine stop".

A power lever switch which closes below a PLA of 33 degrees, energizes the Ground Beta Enable Solenoid (GBES), if the blade angle is less than a predetermined minimum and the aircraft is on the ground. A detent on the power lever quadrant stops unintentional movement of the lever below flight idle during flight.

When the GBES is energized, a pilot valve vents the chamber at the end of the Ground Beta Enable Valve spool (GBEV) to drain. The spring at the other end moves to its ground position. Fine pitch oil pressure then enters another chamber in the PCU, which allows for ground beta blade angles down to reverse. Failure of the GBEV spool to move to its in−flight position (causing loss of overspeed protection) can be proved by doing an OSG test.

Actual engine power is set as a function of rating, air inlet temperature, inlet pressure, aircraft velocity, compressor bleed selection and power turbine speed. Maximum power selected either on the Engine Control Panel or power lever, gives 100% Np irrespective of Np selected by the condition lever.

Three discrete speeds 850, 900, and 1020 rpm are set by the condition lever. When the power lever is in the beta range, propeller speed is usually governed by the FADEC and engine fuel system at 60% RPM.

NOTE

Propeller speed protection on the ground is supplied by the engine, since speed is engine driven rather than airspeed driven.

FADEC overspeed protection may operate in flight, but is effective only in limiting overspeed due to runaway of the engine fuel governor. The FADEC controls fuel flow to the engine to protect the engine and propeller from high torque conditions that would result from an inadvertent propeller feather.

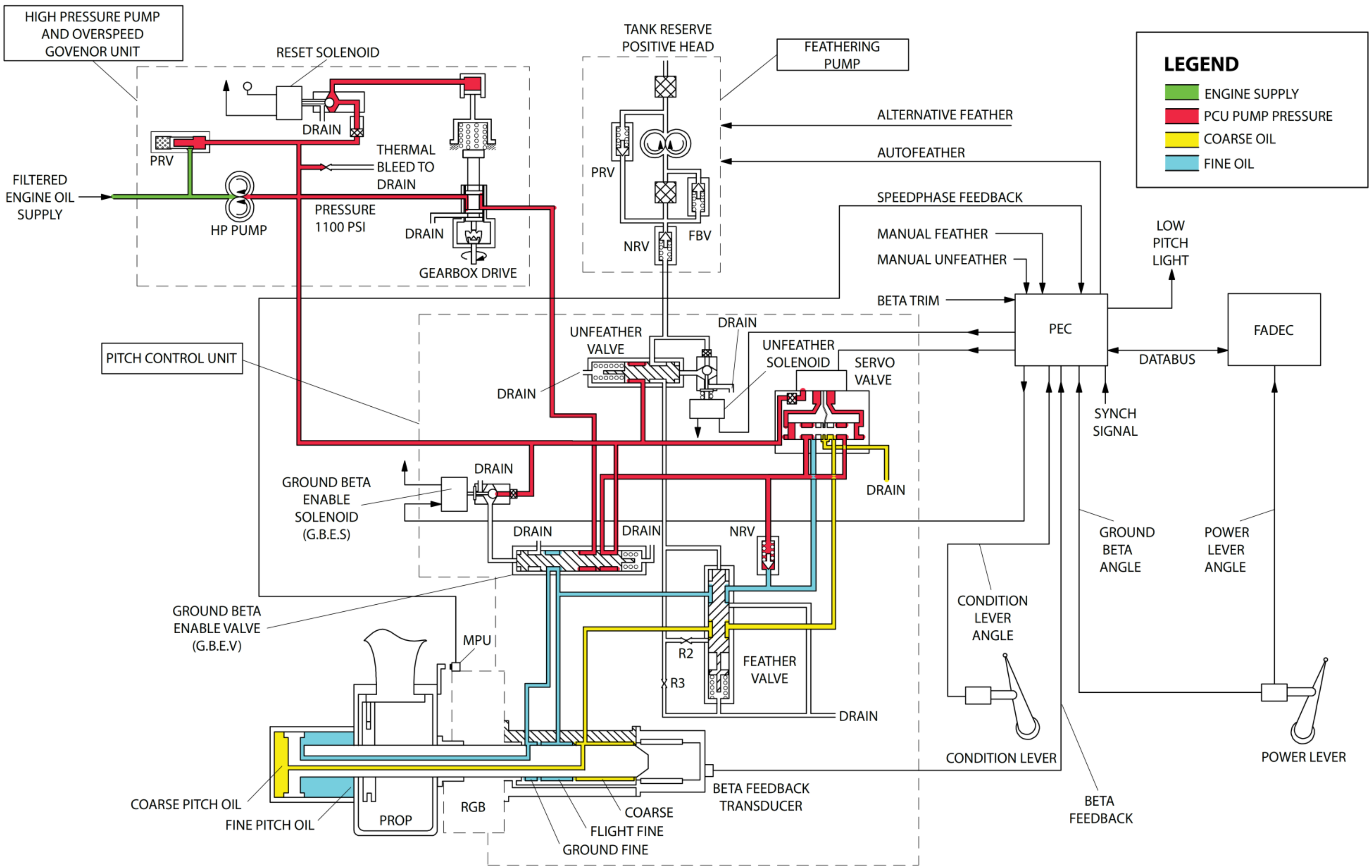


Figure 61-17 Ground Beta Control Mode

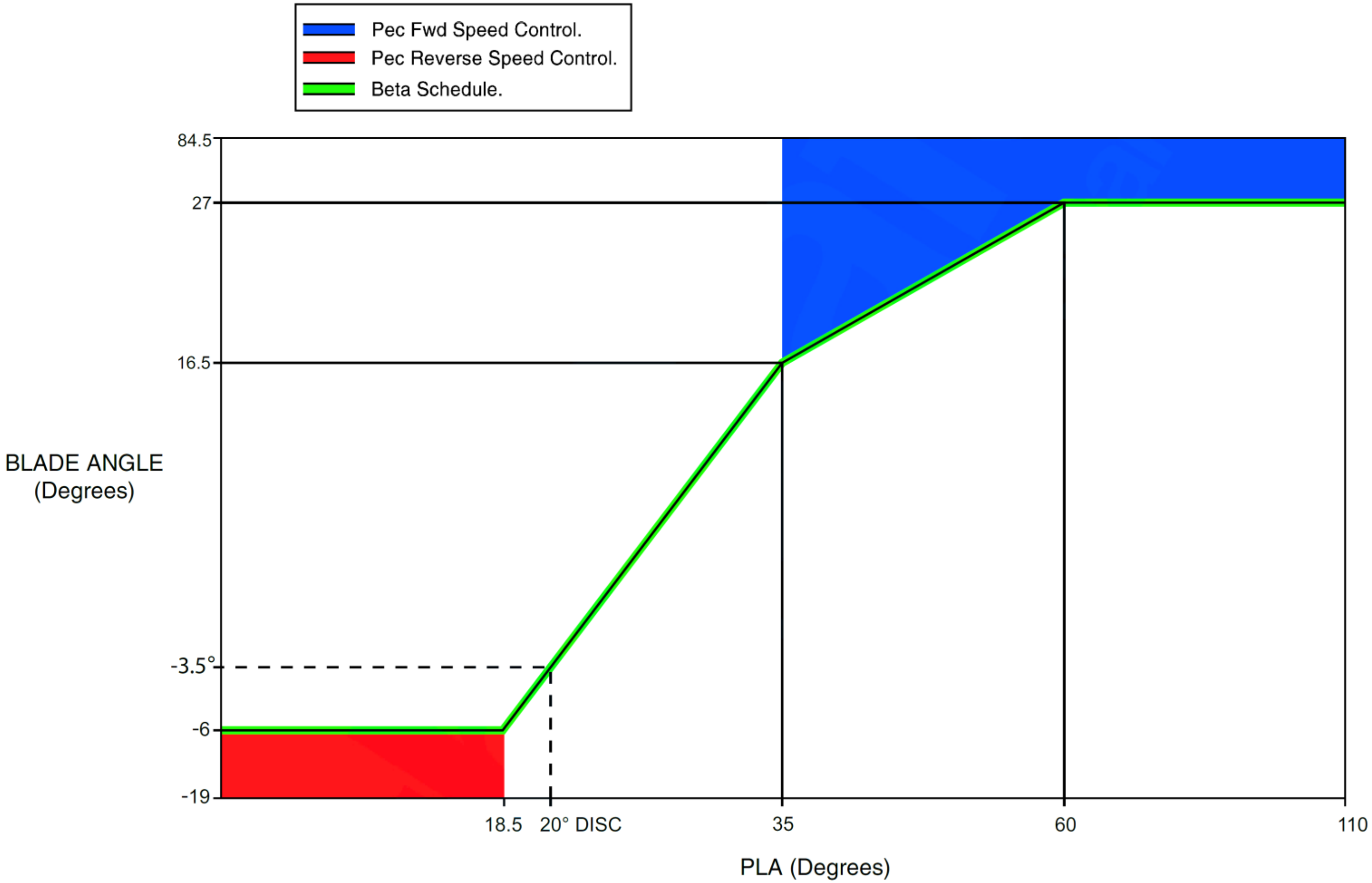


Figure 61-18 Beta Schedule

Reverse Speed Control

Refer Figure 61-19

In this mode the system operates in closed loop propeller RPM control to maintain the propeller speed between 660 and 950 RPM. The engine schedules fuel based on a power schedule versus PLA, with a maximum limit of 1500 SHP. At low airspeeds it is possible that propeller could reach the maximum reverse stop, the propeller rotational speed is then controlled by the engine OSG and can increase up to 1020 RPM.

This mode is similar to forward speed control, except that it operates in reverse, i.e driving more negative to absorb more power and reduce propeller RPM.

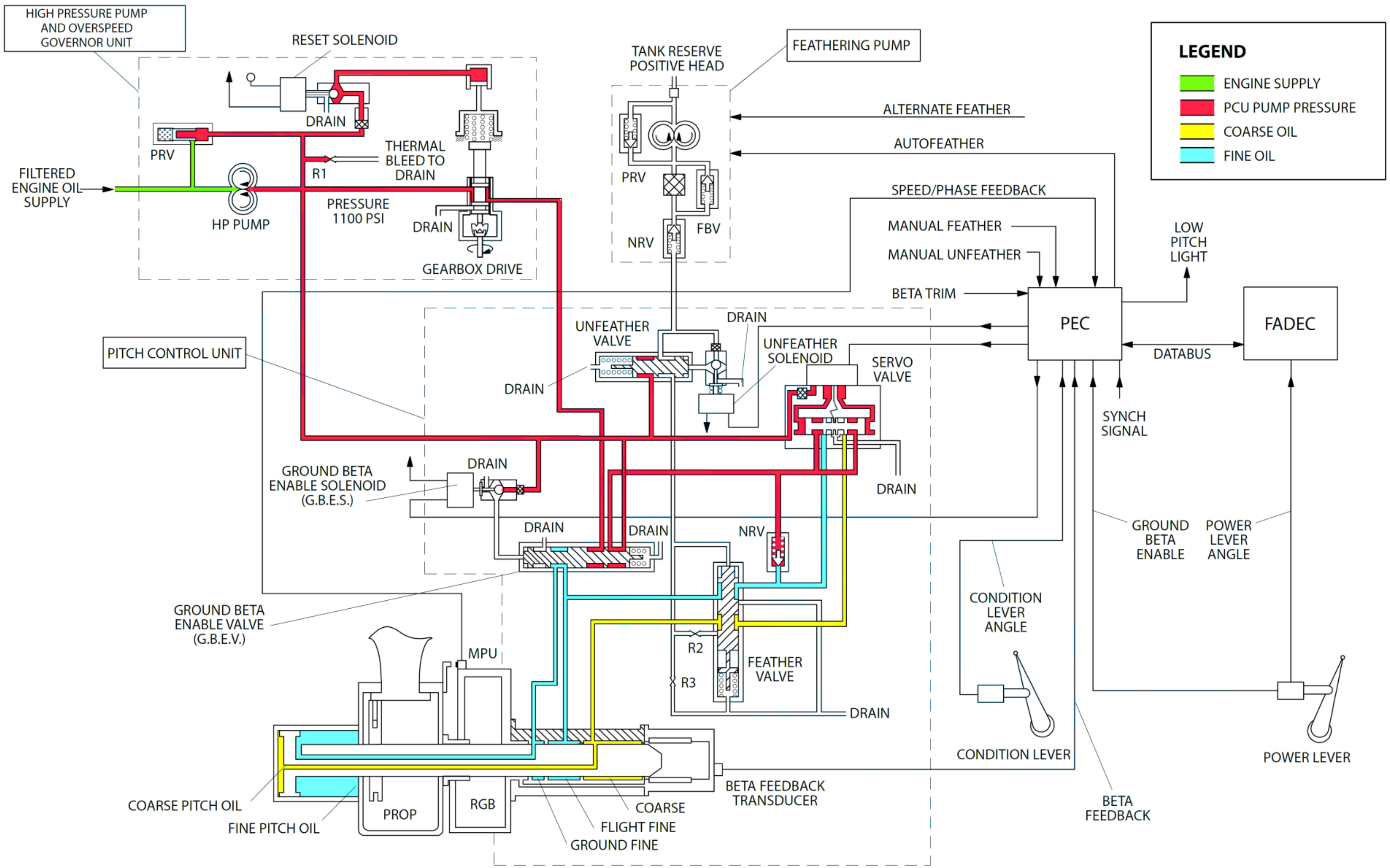


Figure 61-19 Maximum Reverse

Synchro-phase Control

Refer Figure 61-20

In this mode the system operates in closed loop propeller speed and propeller phase control. Synchrophasing acts to reduce cabin noise, by making sure the phase difference between the slave propeller and the master propeller is controlled to a demanded angle.

The phase angle is calculated by timing the differences between the master and slave propeller MPU signals over a complete propeller revolution. The phase demand is controlled from the condition lever position. Synchrophasing will give phasing accuracy of better than plus or minus 1 degree.

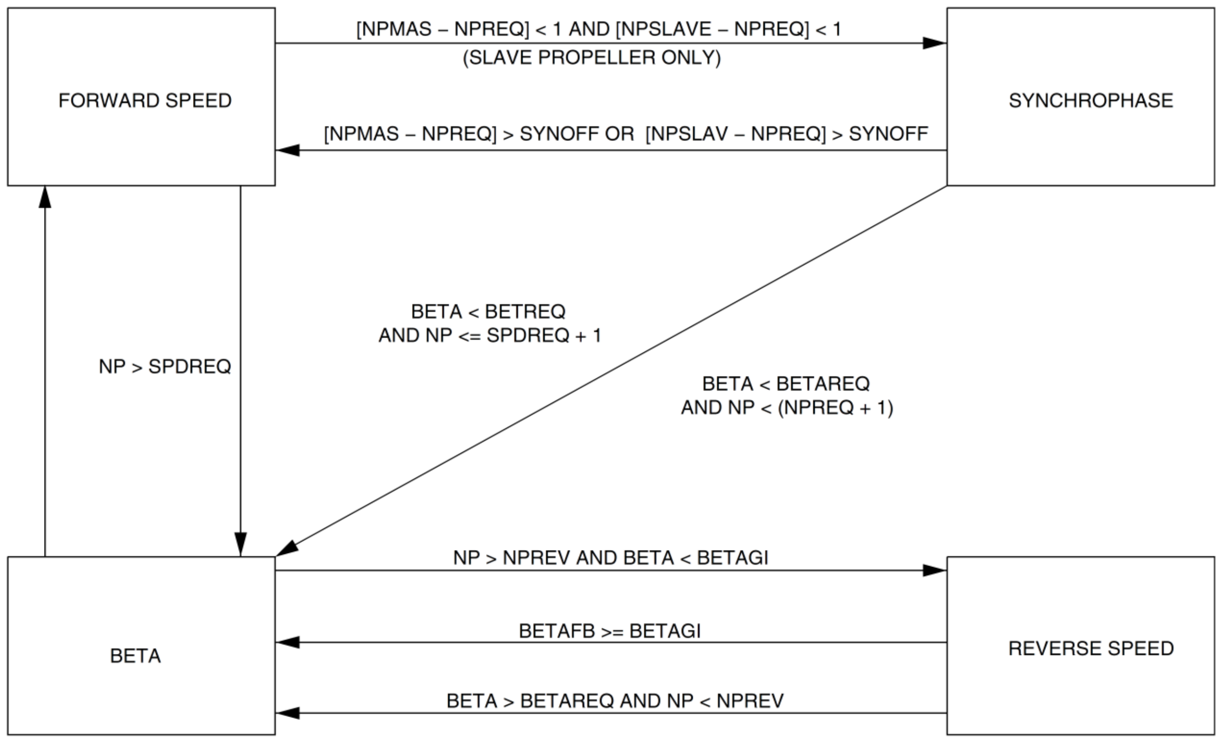


Figure 61-20 PEC Control Mode Transitions Including Synchrophase

Normal Feather

Refer Figure 61-21

The purpose of the normal feather mode is to feather the propeller on routine engine shutdowns.

Routine feather is set when the condition lever is moved to either the START/FEATHER or FUEL OFF detent. A routine feather signal commands the PEC to drive the servo valve towards coarse pitch.

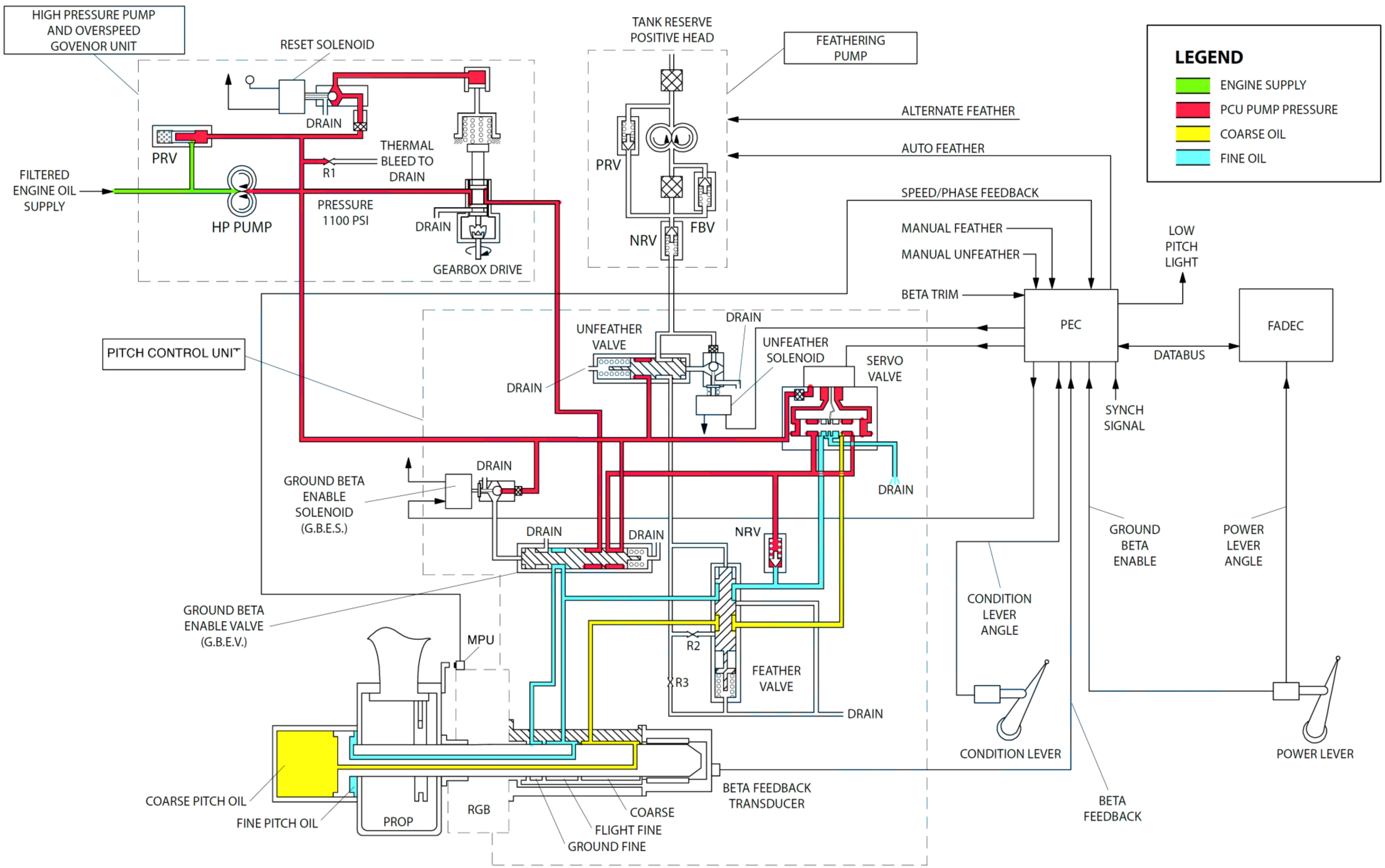


Figure 61-21 Normal Feather

Autofeather

The purpose of the autofeather system is to:

- Reduce drag on the failed engine
- Increase power on the serviceable engine
- Prevent the two engines from going into autofeather simultaneously

Refer Figure 61-22

The autofeather function is implemented in PEC hardware. It includes cross-wing communication with the remote PEC.

Two torque signals from the NPT/Q sensors on an engine are need to show less than 25% torque for three seconds before the system autofeathers.

The autofeather function is armed by:

- PLA to above 60°
- Local and remote torques above 50%
- Pressing the AUTOFEATHER SELECT switch.

When both powerplant Autofeathers are armed, an A/F ARM indication is shown on the ED.

Autofeather is disarmed by either de-selecting the AUTOFEATHER SELECT switch, or moving either PLA below 60.

An autofeather will cause the PEC to:

- Output a servo-valve drive coarse signal
- Energize the auxiliary pump relay.

At this point the A/F ARM indication is removed from the ED.

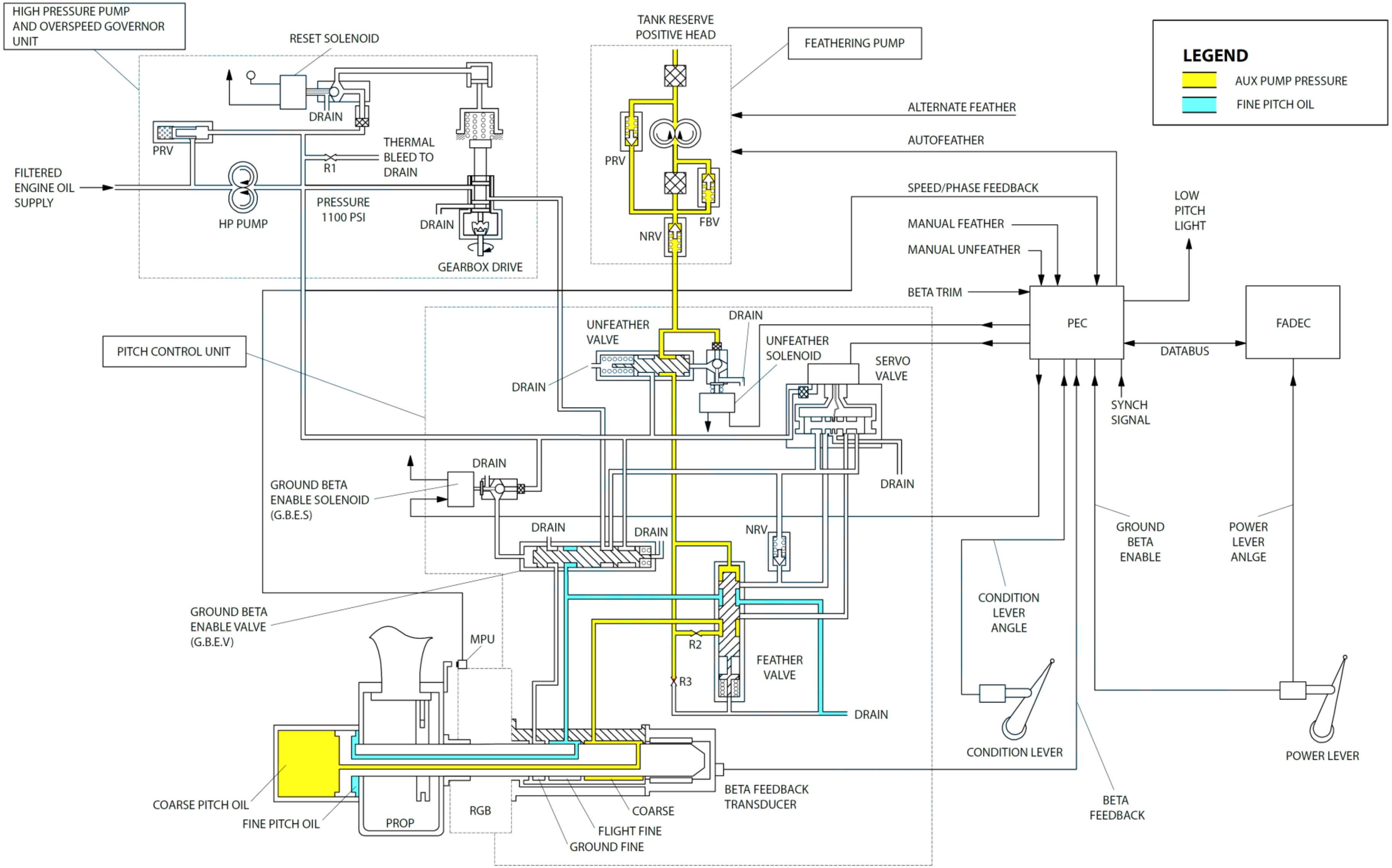


Figure 61-22 Auto Feather Mode

Alternate Feather

The alternate feather is set when the condition lever is moved to the START/FEATHER or FUEL OFF detent and the applicable ALT FTHR switchlight, on the PROPELLER CONTROL panel, is pushed. The auxiliary pump starts and supplies pressure to enable the Back−up Feather Valve and drive the propeller to coarse pitch. The electrical circuit for the alternate feather mode has a switch located on the condition lever. The switch closes at, or below the START/FEATHER position, to make sure that the flight crew must try the normal feather mode before selecting alternate feather. The alternate feather function gives an independent method of feathering the propeller and can override the authority of the PEC.

Maintenance Unfeather

On the ground, a static propeller can be unfeathered, by selecting the applicable MAINTENANCE UNFEATHER switch on the maintenance panel to ON. This energizes the auxiliary pump and the unfeather solenoid, scheduling oil pressure to the servo valve. The PEC controls the servo valve output to drive the propeller towards fine pitch if all of the following are true:

- There is no propeller rotation
- Maintenance mode is selected
- Weight−On−Wheels input signals are detected.

Automatic Underspeed Protection Circuit (AUPC)

Refer Figure 61-23

The AUPC prevents the feathering of the two propellers (loss of thrust) to protect against a common software error in PEC1 and PEC2. This would result in a drive coarse signal on both propellers and loss of thrust on take−off. AUPC function is implemented in hardware independently of the PEC control lane software. In the event of an underspeed being detected on both Npt sensors, a drive fine signal is generated and the Np will increase. The propeller speed increase will be arrested by the Overspeed Governor (OSG).

This function overrides the authority of the control lane software but is, in turn, overridden by the autofeather function.

The propeller underspeed governor (AUPC) is armed as follows:

- Power lever (PLA) is set to more than flight idle
- Condition lever (CLA) is set to more than START & FEATHER
- No manual feather selection
- No autofeather.

AUPC is activated when the conditions that follow are set for more than 1 second:

- Propeller speed (NP) is less than 816 RPM
- Torque (TRQ) is over 50%
- CLA is selected to START/FEATHER in flight, then selected to speed governing with the engine power at cruise or higher.

The AUPC activation disables the PEC speed governing and beta control and sends an unmodulated drive fine signal to the PCU.

AUPC activation is indicated as follows:

- Propeller pitch decreases
- Propeller speed increases to 1060/1070 rpm and the propeller speed stays at this value unless the propeller is feathered or the blade angle goes to the fine pitch stop during the landing
- PEC caution lights come on
- POWERPLANT advisory lights come on in the engine display.

NOTE

These indications are latched and the propeller will not unfeather until reset by electrical power interruption of the PEC.

AUPC is disarmed as follows:

- PLA moved below FI
- CLA moved to Start/Feather
- Autofeather or manual feather demanded.

The AUPC activation will cause the propeller to go to the flight fine stop or to overspeed governing if the airspeed is high.

The AUPC function is tested during Autofeather Test.

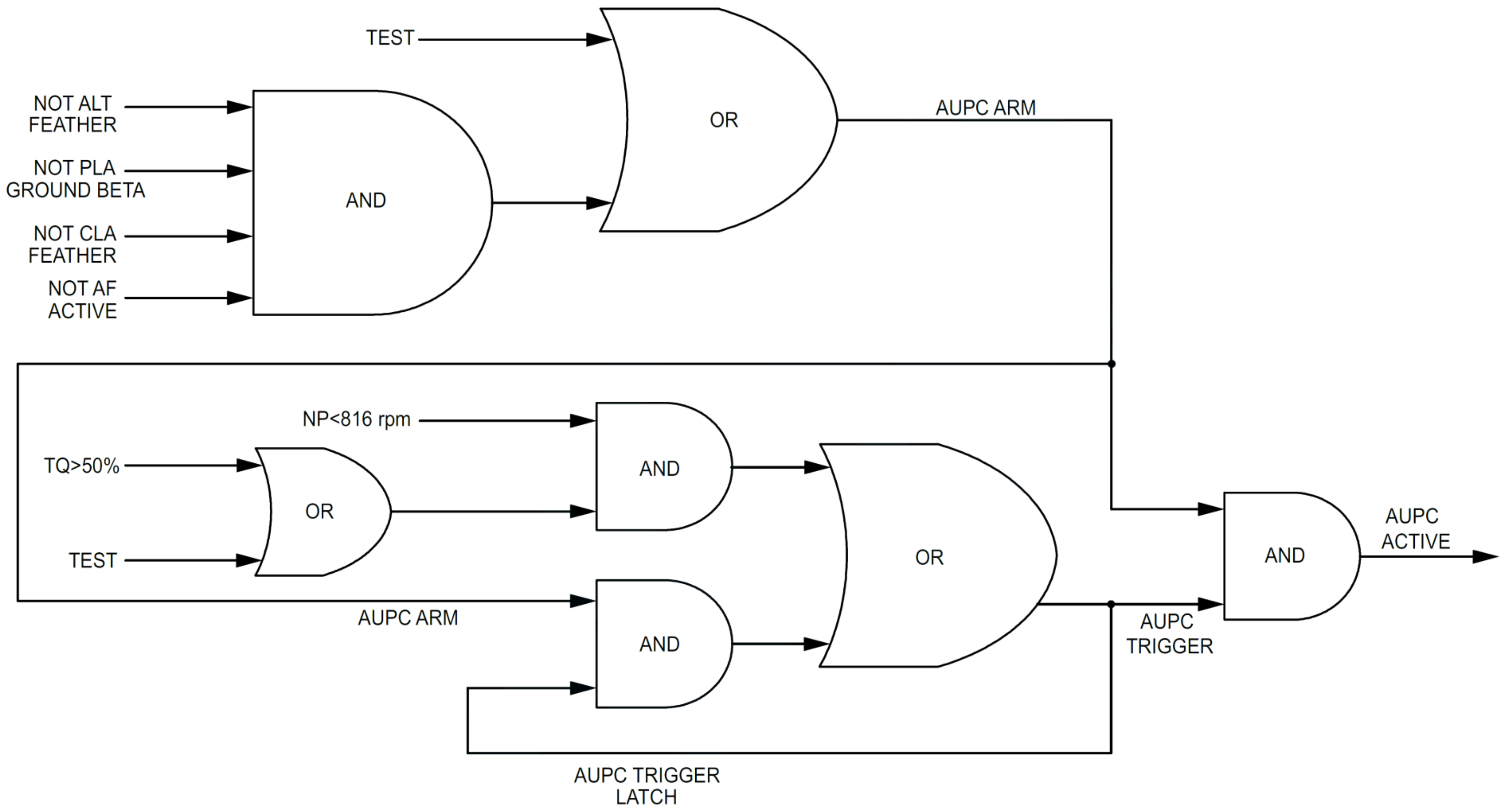


Figure 61-23 AUPC Logic

Automatic Take-Off Thrust Control System (ATTCS)

The purpose of the ATTCS is to give the necessary protection against engine failure during the critical part of the take-off roll.

The ATTCS causes an uptrim of the remote powerplant and an auto-feather on the failed powerplant.

Any erroneous feather of the local propeller will cause an uptrim of the remote powerplant.

The ATTCS has two sub-systems:

- Autofeather
- Uptrim

Autofeather

Refer Figure 61-24

The autofeather function is implemented in PEC hardware and includes cross–wing communication with the remote PEC, to prevent both powerplants autofeathering simultaneously. Two torque signals are needed to show less than 25% torque for three seconds before the system autofeathers.

Autofeather is armed when both the engine torques exceed a minimum value of 50%, both the power levers are set to more than 60 degree PLA and the AUTOFEATHER SELECT switch is selected ON. When both powerplants are armed, an A/F ARM indication is shown on the Engine Display (ED).

Autofeather is disarmed by either de–selecting the AUTOFEATHER SELECT switch, or moving both PLAs below the rating detent.

An autofeather will cause the PEC to:

- Output a servo valve drive coarse signal
- Energize the auxiliary pump relay.

At this point the A/F ARM indication is removed from the ED.

When the propeller is autofeathered, the system can only be disarmed by either, de–selecting the AUTOFEATHER SELECT switch, or moving both PLAs below the rating detent.

A detected failure in the ATTCS results in the system inhibiting the A/F ARM indication.

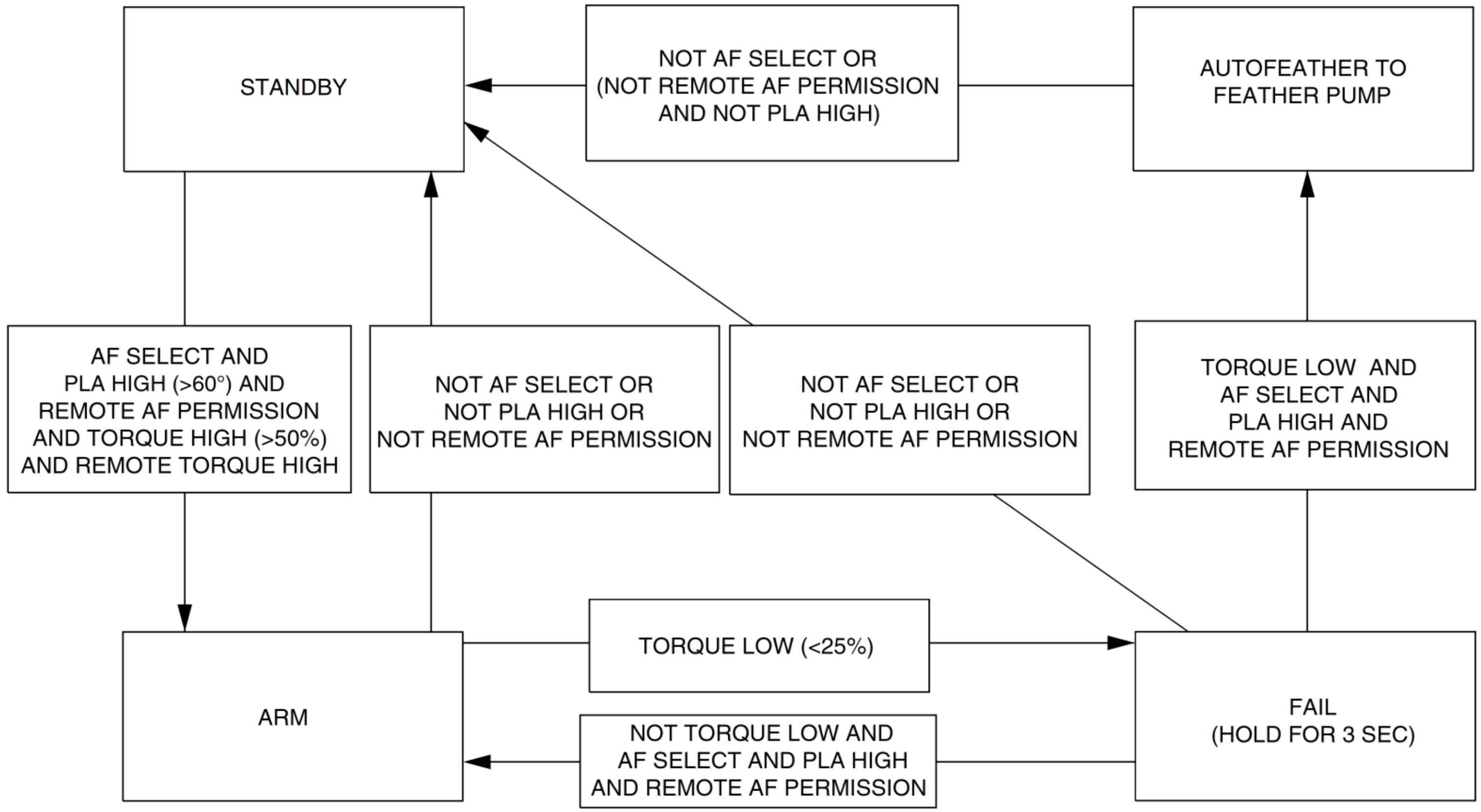


Figure 61-24 Autofeather State Transitions

Uptrim

Refer Figure 61-25

The purpose of the Uptrim system is to uptrim the local engine by 10%, when signalled to do so by the remote PEC.

The uptrim function is implemented in the PEC hardware, with communication from the local PEC to the remote FADEC.

Uptrim is armed by moving both PLA above 60° and having remote torque high (Q) for > two seconds.

Uptrim is disarmed by moving either PLA below 60° or by autofeather of the remote powerplant.

When the system is armed an uptrim will occur if:

- Both torque signals drop below 25%

or;

- Both torque sensors show a speed of less than 800 N_p

or;

- Autofeather active

An uptrim will cause the PEC to command the FADEC to change from the NTOP schedule, to the Maximum Take-Off Power schedule.

An UPTRIM message will appear on the ED.

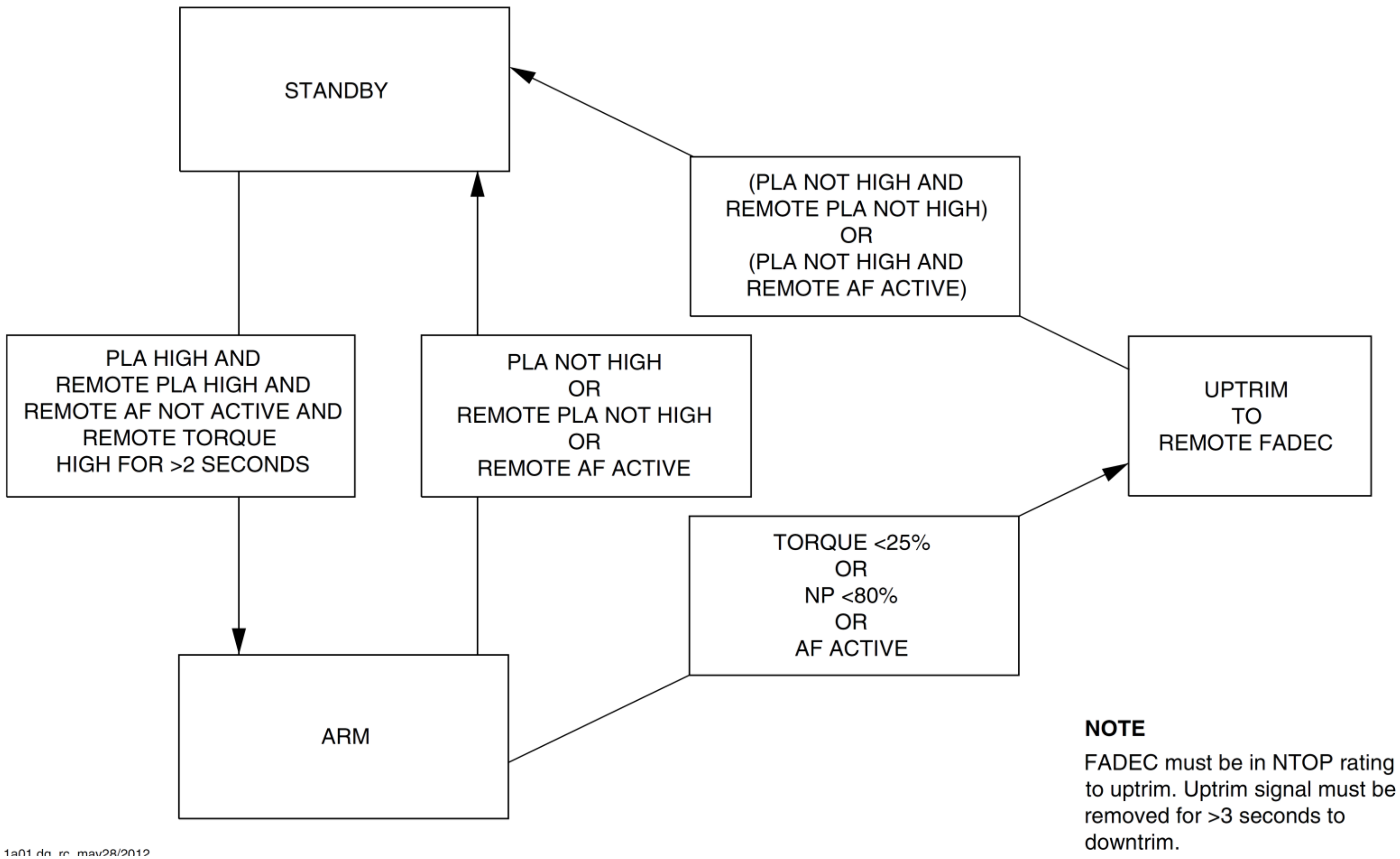


Figure 61-25 Uptrim for Low Np or Low Torque

Alternate Feather and PROPELLER CONTROL panel

Refer Figure 61-26

The purpose of the Alternate Feather system is to give an alternate means to feather the propeller of a failed engine.

Alternate feather is set when the condition lever is moved to the START/FEATHER, or FUEL OFF position, operating the CLA < 40°, or the microswitch and the applicable ALT FTHR switchlight, on the PROPELLER CONTROL panel is pushed.

The auxiliary pump starts (30 second run time) and supplies pressure to enable the back-up feather valve and drive the propeller to coarse pitch.

The control panel is installed on the upper center console in the flight compartment.

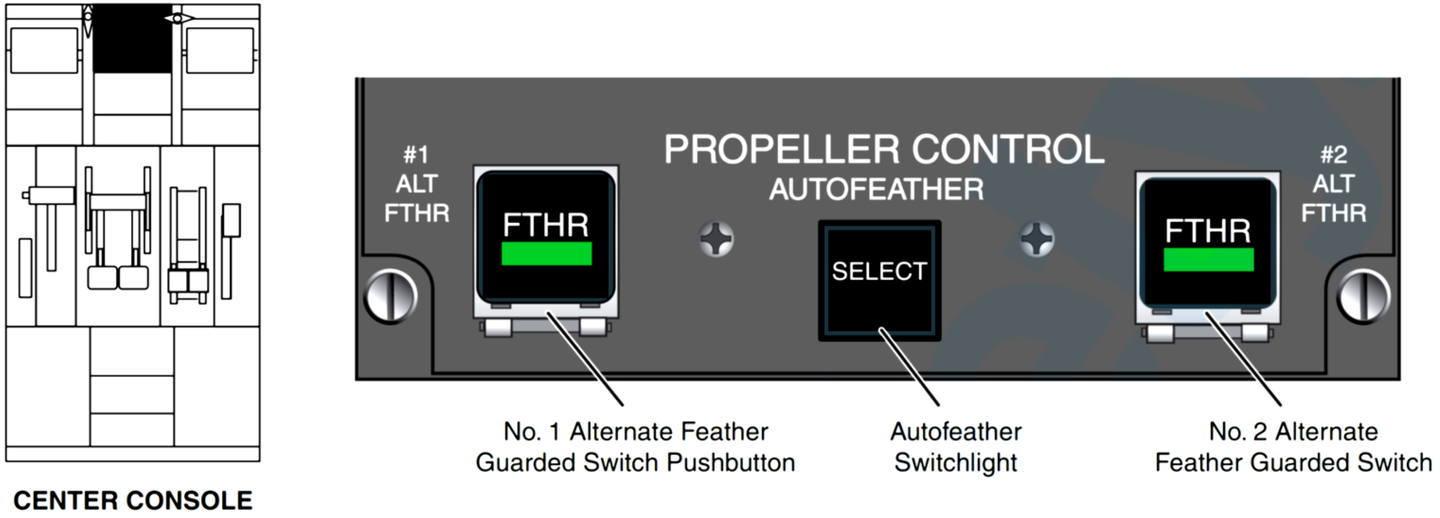


Figure 61-26 Propeller Control Panel

Propeller Control – Component Description

Beta Tube Assembly

Refer Figure 61-27

The beta tubes are installed in the center of the propeller hub.

The assembly is attached to the front of the crosshead assembly by means of an adjustable screw thread and locking collar with a spline. The adjustable screw permits setting of the assembly within the Beta Feedback Tube to achieve full range of feedback.

The beta tubes are used to transfer fine and coarse pitch oil pressure from the PCU to the propeller pitch change mechanism. The tubes also monitor the blade pitch angle for beta control.

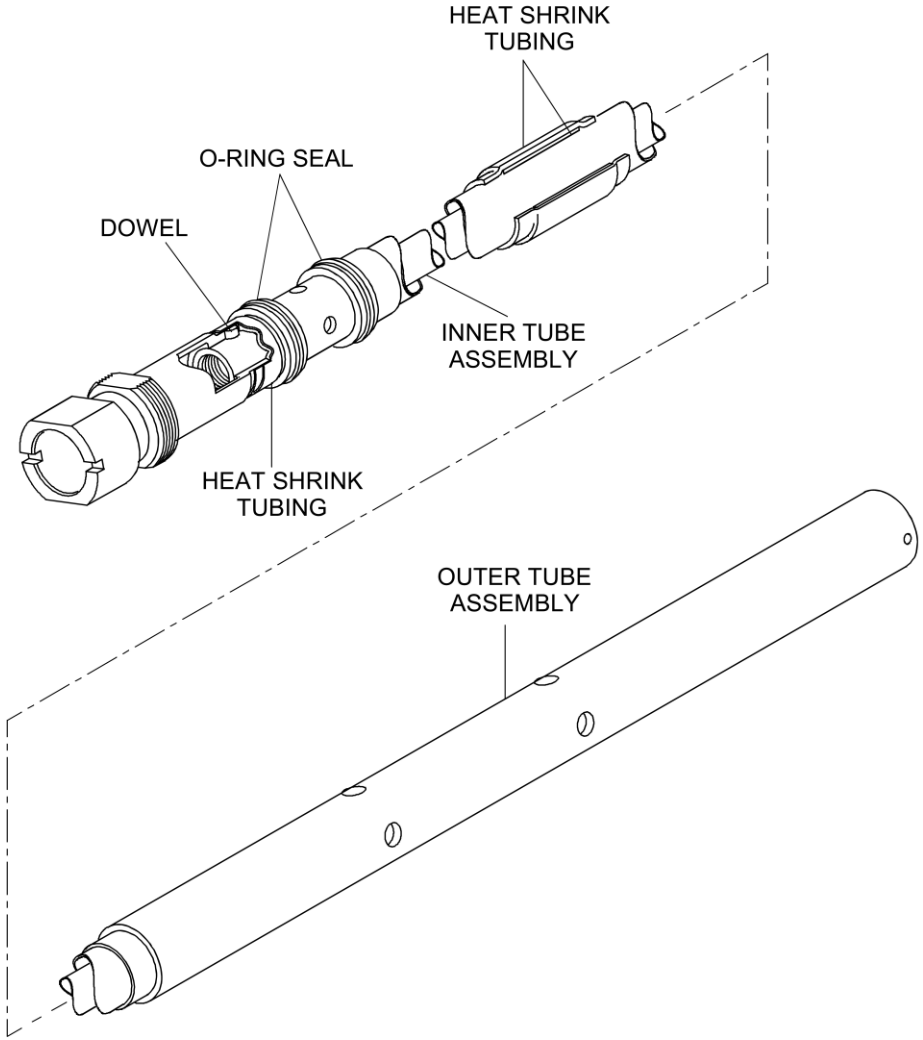


Figure 61-27 Beat Tube Assembly

Magnetic Pickup Unit (MPU)

The speed and phase of the propeller is sensed by a Magnetic Pickup Unit and a set of 7 targets on the deicing slip ring. Six targets give speed signaling inputs and one acts as a master reference for balancing purposes.

The MPU is a dual channel device located on the front of the reduction gearbox.

Pitch Control Unit (PCU)

Refer Figure 61-28

The PCU controls the flow of oil pressure to the fine and coarse pitch sides of the propeller pitch change mechanism. This is done by the modulation of high pressure oil supplied by the Overspeed Governor (OSG) using a servo valve. The servo valve receives electrical signals from the PEC and directs oil pressure through a transfer sleeve and beta tubes, to the coarse/fine sides of the propeller pitch change piston.

The PCU has the components that follow:

- Servo valve
- Ground Beta Enable Solenoid Valve
- Unfeather valve
- Back up Feather Valve
- Beta Feedback Transducer

The PCU rear case gives the interface with the beta feedback transducer (BFT). The Ground Beta Enable Solenoid Valve (GBEV), the Unfeather Solenoid Valve, the servo valve, and the Beta Feedback Transducer (BFT), are not line replaceable units. The PCU is located on the rear face of the reduction gearbox and is attached to the gearbox by a V−band clamp.

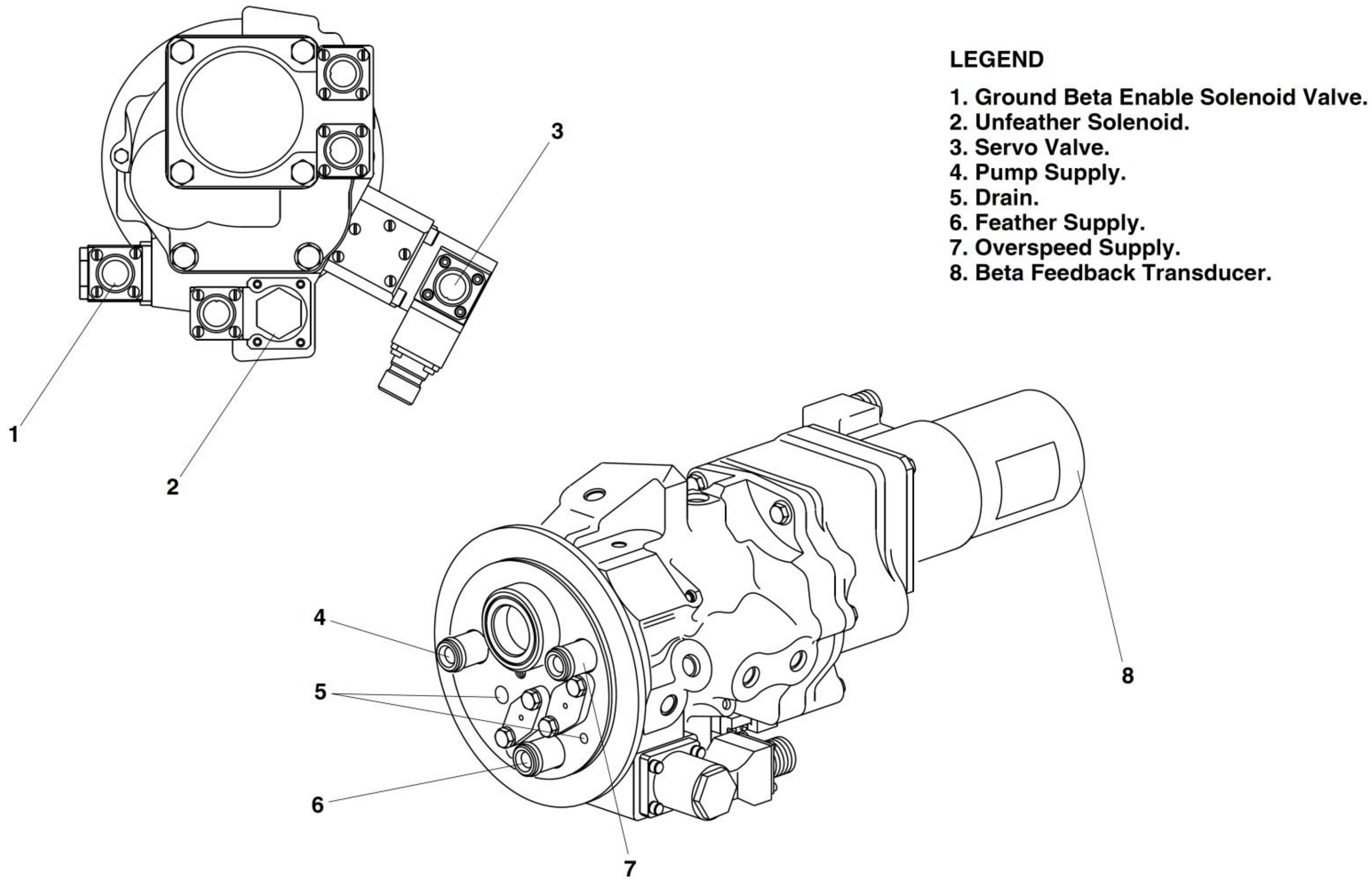


Figure 61-28 Pitch Control Unit

Servo valve

Refer Figure 61-29

The servo valve is a two stage nozzle flapper design, which is used to control blade pitch in all control modes. The torque motor drive on the first stage has two coils, connected to each PEC lane. Opening the valve schedules high pressure oil to one line, while venting the other line to drain. The valve is biased such that the propeller is driven towards coarse pitch in the event of an interruption of the electrical drive signal from PEC.

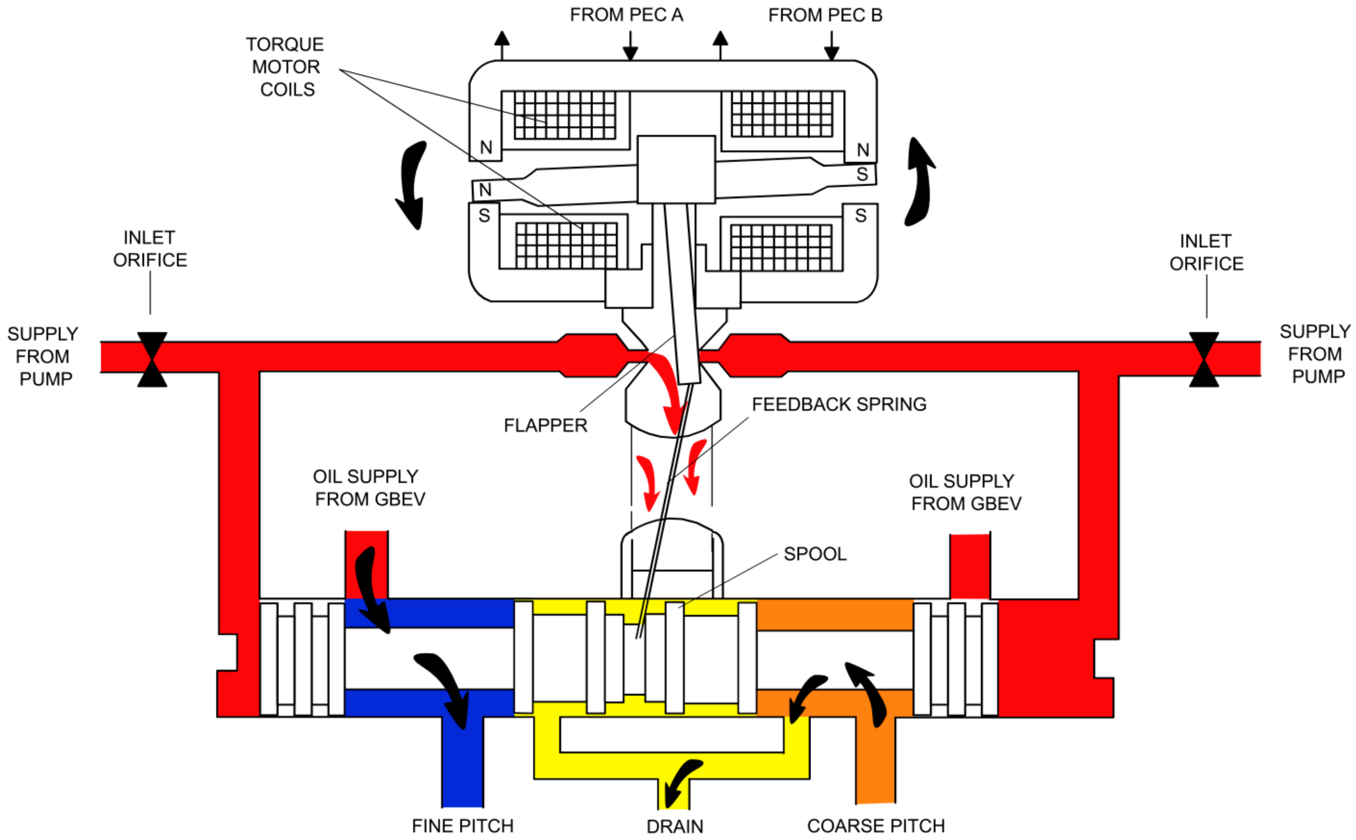


Figure 61-29 Servo Valve

Ground Beta Enable Solenoid Valve (GBEV)

Refer Figure 61-30

The Ground Beta Enable Valve stops the propeller entering the reverse pitch range during flight. The valve is controlled by the Ground Beta Enable Solenoid. When energized it vents the enable valve end chamber to drain letting spring pressure move the valve from the flight position into the ground position.

In the flight position (solenoid de−energized), the ground fine chamber in the PCU is vented to drain. This stops the propeller being driven below the flight fine stop, which is set just below flight idle pitch. The OSG is connected into the hydraulic system with the valve in this position.

In the ground position (solenoid energized), the ground fine chamber is connected to the fine pitch oil line. This lets the propeller be driven into the ground blade pitch range. The OSG is isolated from the system to prevent potential coarse pitch hang ups during selection of maximum reverse after touchdown, or full power from the reverse position. The engine fuel valve controller gives overspeed protection with the valve in this position.

NOTE: For maintenance feathering and unfeathering, where there is no propeller rotation and no output from the high pressure pump, a spring in the Ground Beta Enable Valve makes sure that the valve is in the ground position.

The GBEV is installed in the PCU.

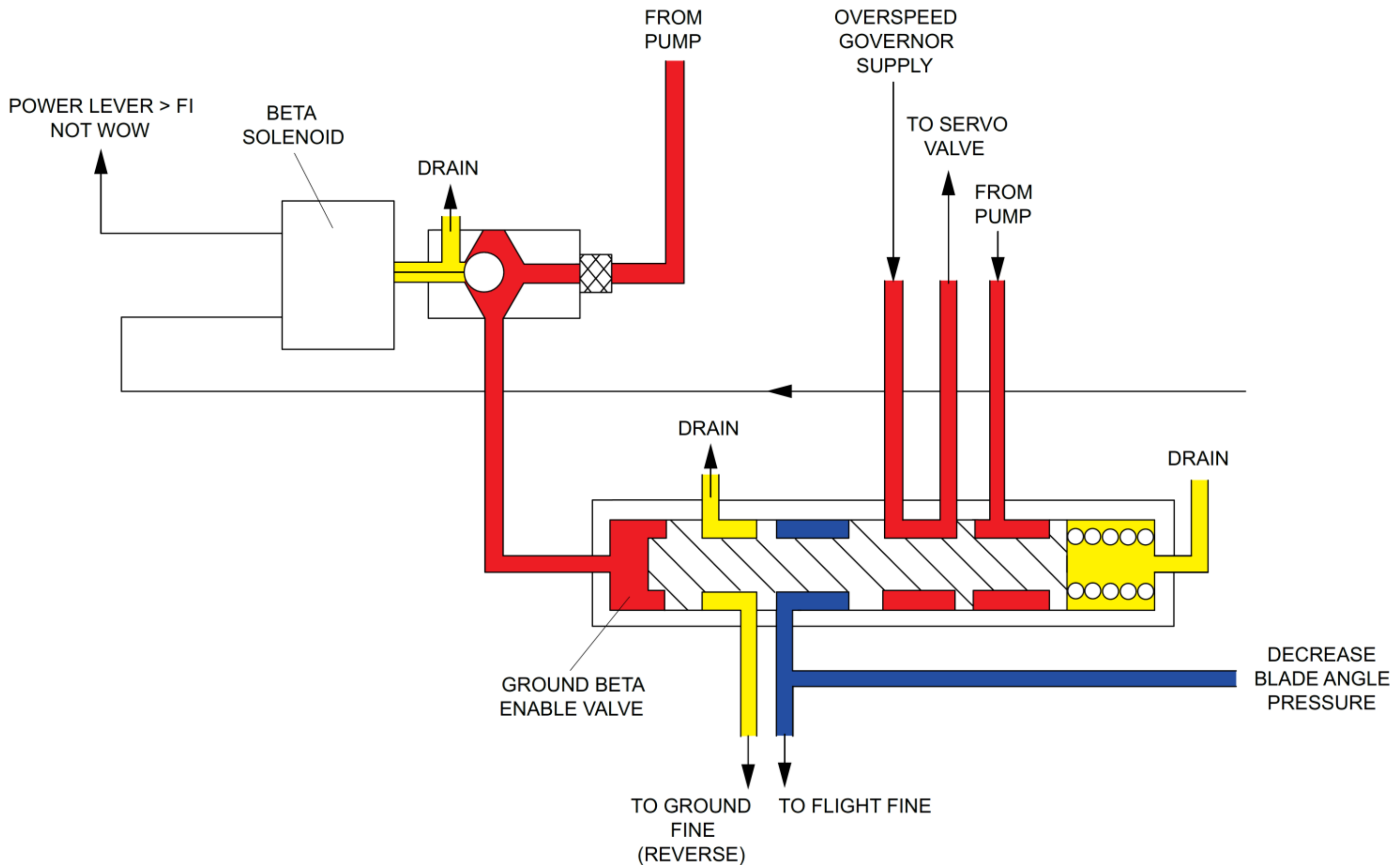


Figure 61-30 Ground Beta Enable Solenoid Valve

Unfeather Valve and Solenoid

Refer Figure 61-31

The purpose of the Unfeather Valve and Solenoid is to direct oil from the auxiliary pump to the PCU servo-valve.

The valve is enabled when the unfeather solenoid and auxiliary pump are energized.

The unfeather solenoid is activated by a maintenance switch, and is monitored by the PEC, although the PEC has no control over the valve.

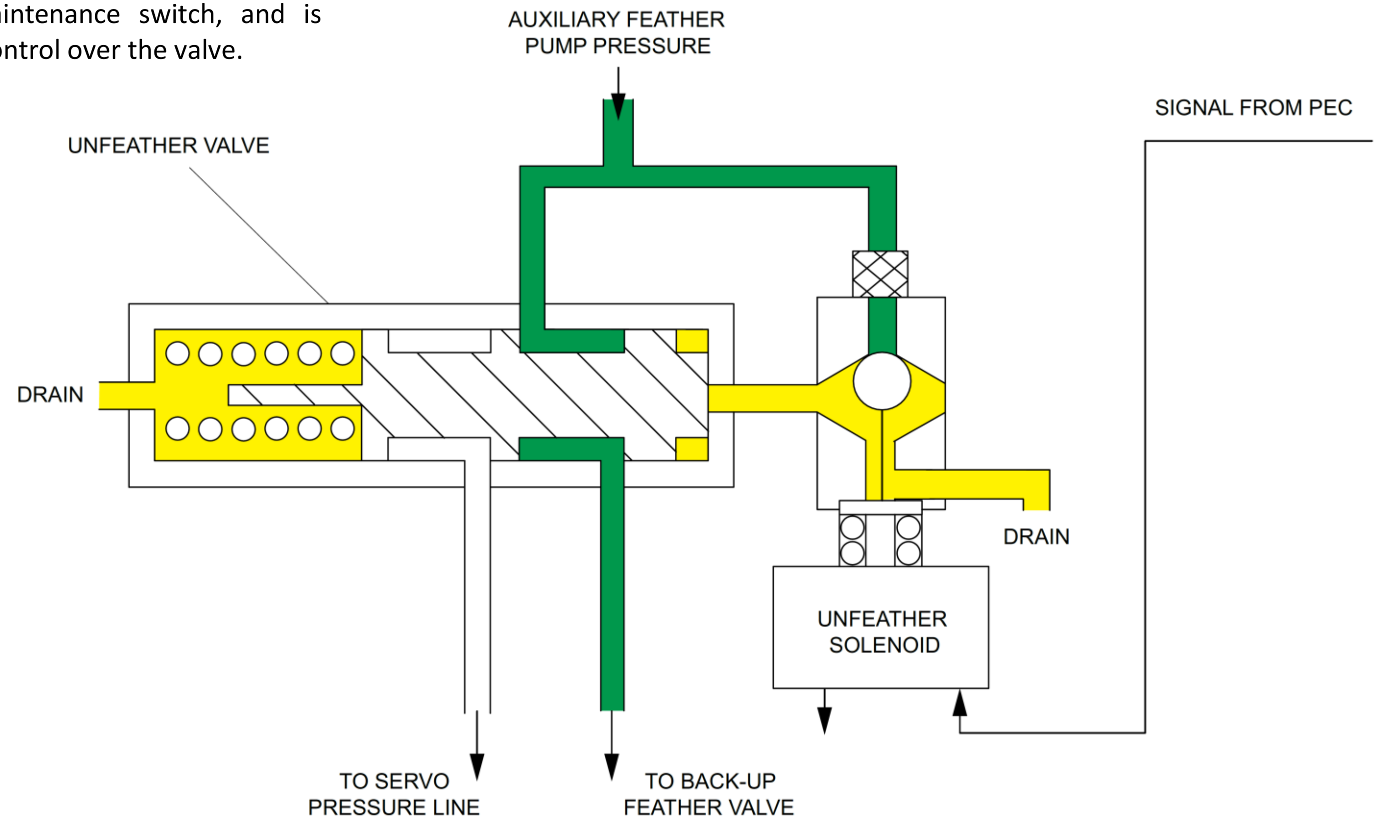


Figure 61-31 Unfeather Valve and Solenoid

Beta Feedback Transducer (BFT)

Refer Figure 61-32

The Beta Feedback transducer is mounted on the PCU, at the back of beta tube assembly. The Beta Feedback transducer monitors the blade pitch angle for beta control.

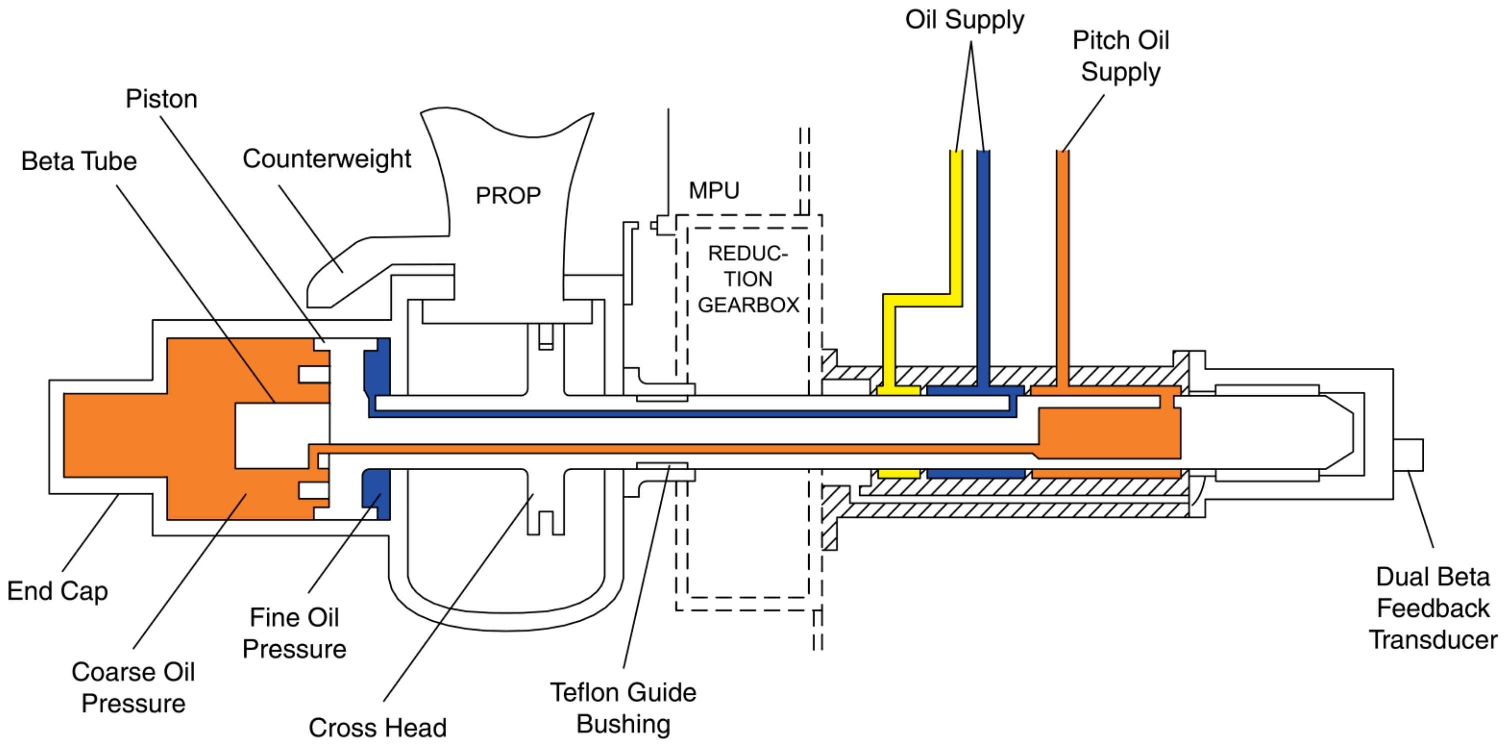


Figure 61-32 Beta Feedback Transducer

Overspeed Governor and High Pressure Pump (OSG)

Refer Figures 61-33

The unit has two primary functions:

- Increase the pressure of engine supplied oil, to a pressure suitable for propeller actuation
- Control the supply of oil to the propeller during flight.

If an overspeed occurs, at approximately 104% Np, the oil supply from the PCU will be vented to drain. This lets the propeller move towards coarse pitch by action of the blade counterweights, reducing Np. The unit has a reset solenoid which sets the governor to a lower value, approximately 80% Np, when doing functional checks.

The OSG has a gear pump and fly weight governor driven by the reduction gearbox. The governor controls the position of a spring loaded spool. When an overspeed condition occurs the spool will move against the spring, to vent the PCU supply line to drain.

The unit is a two piece aluminum constructed body housing the gear pump case and governor/spool. The governor/spool body houses the reset solenoid valve and a pressure relief valve which limits pump maximum output pressure to 1100 psi (758.4 kPa). The reset solenoid and pressure relief valves are line replaceable items. The OSG and pump are attached to the reduction gearbox by four studs and nuts. Correct installation is assured by one dowel. A bonded seal plate provides sealing against the gearbox casing for four oil ports.

On Post SB 84−61−02 overspeed governors, the seal plate is replaced with four O−rings which install into seal grooves in the face of the pump adapter of the OSG.

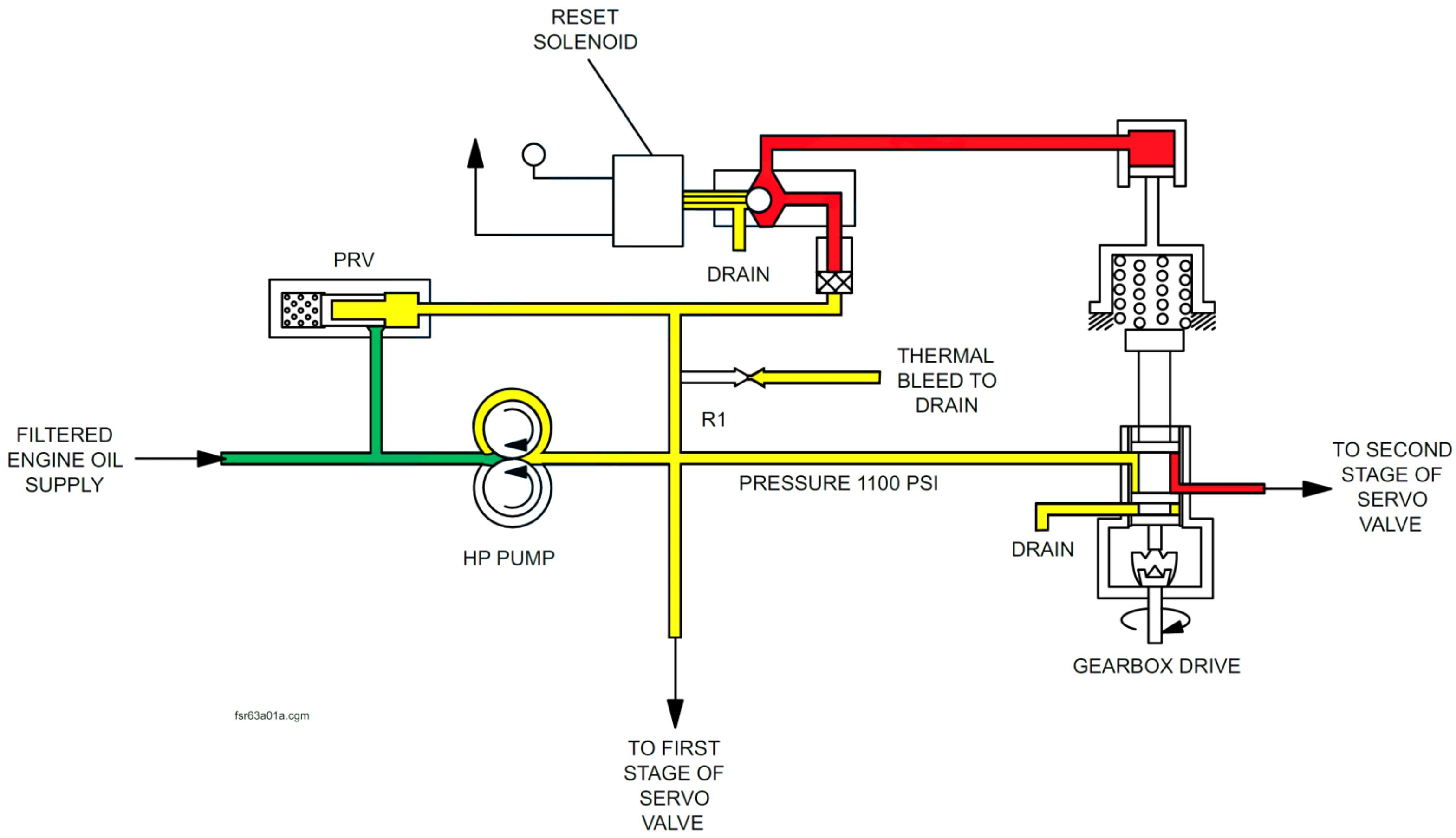


Figure 61-33 Propeller Overspeed Governor and High Pressure Pump

Feathering Pump

Refer Figures 61-34

The feathering pump feathers the propeller independently of the normal control system and achieves full feather (zero wind milling). The pump supplies feathering or unfeathering oil pressure if the engine supply is not available.

It gets its oil supply from the auxiliary oil tank in the reduction gearbox (refer to SDS 79−20−00). The feathering pump is the only supply from the bottom of the auxiliary oil tank (all other supplies are from the top), which reserves the tank volume for the feathering pump when the engine is not running. When energized, the pump supplies unmodulated oil pressure to the PCU, which will feather or unfeather the propeller blades.

The pump is energized during:

- Autofeather
- Alternate feather
- Maintenance unfeather

The unit has a gear pump and 28V DC electric motor. The electric drive to the pump has a run time protection relay to prevent damage to the unit. The pump has an aluminum body that contains a gear pump and a reduction gearbox, a filter, filter bypass valve, and a pressure relief valve.

The pump is located on the front case of the reduction gearbox and is attached by four studs and nuts. Two oil transfer bobbins and "O" ring seals give the sealing for the supply and outlets ports of the pump.

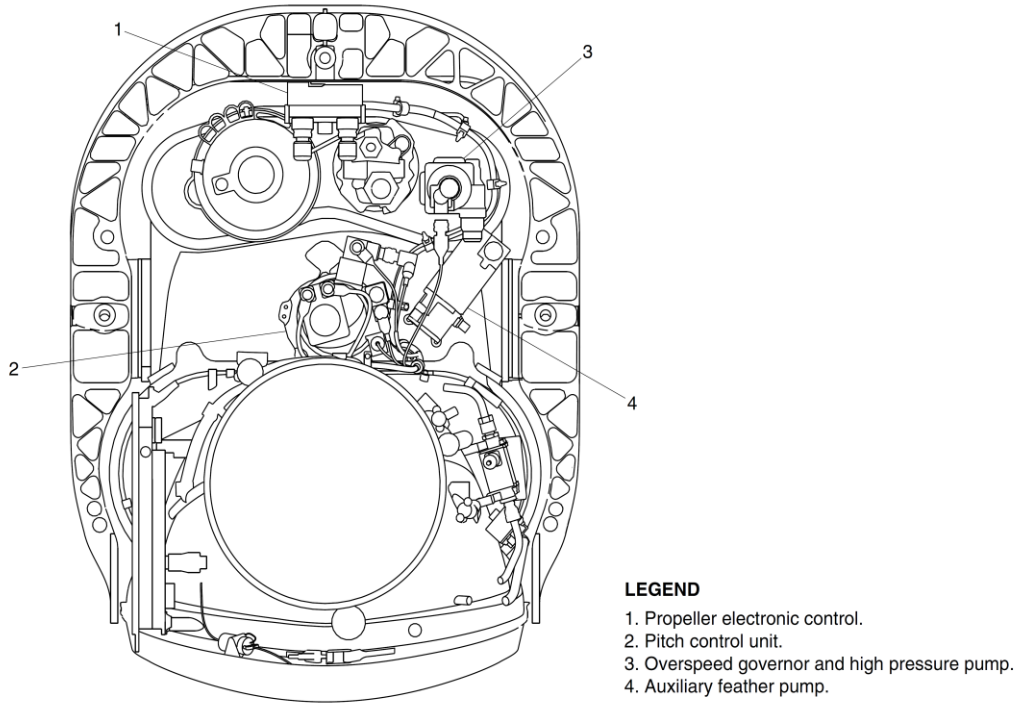


Figure 61-34 Feathering Pump Location

Propeller Electronic Control Unit (PEC)

Refer Figures 61-35 & 61-36

The Propeller Electronic Controller (PEC) controls the various modes of the propeller system. The modes are as follows:

- Speed control in−flight
- Ground beta control
- Reverse thrust speed control
- Autofeather/Engine uptrim command
- Synchrophasing

The PEC also does numerous BITE and pilot initiated tests and has comprehensive fault management software. The unit has two control lanes, one active and one on standby, with automatic transfer to the standby lane in the event of a fault. An autofeather module is also installed, and is independent of engine control (FADEC) and control lane modules in the PEC.

The PEC does the functions that follow:

- Control mode and Control function selection
- Servo valve control
- Ground Beta Enable and Overspeed Governor reset solenoid control
- Unfeather solenoid monitoring
- Low pitch indication
- Fault management:
- Fault detection
- Fault accommodation
- Fault storage
- Fault annunciation
- Automatic underspeed protection circuit (AUPC)
- Automatic Takeoff Thrust Control System

PEC output is isolated when both lanes are faulty or all electrical power is lost.

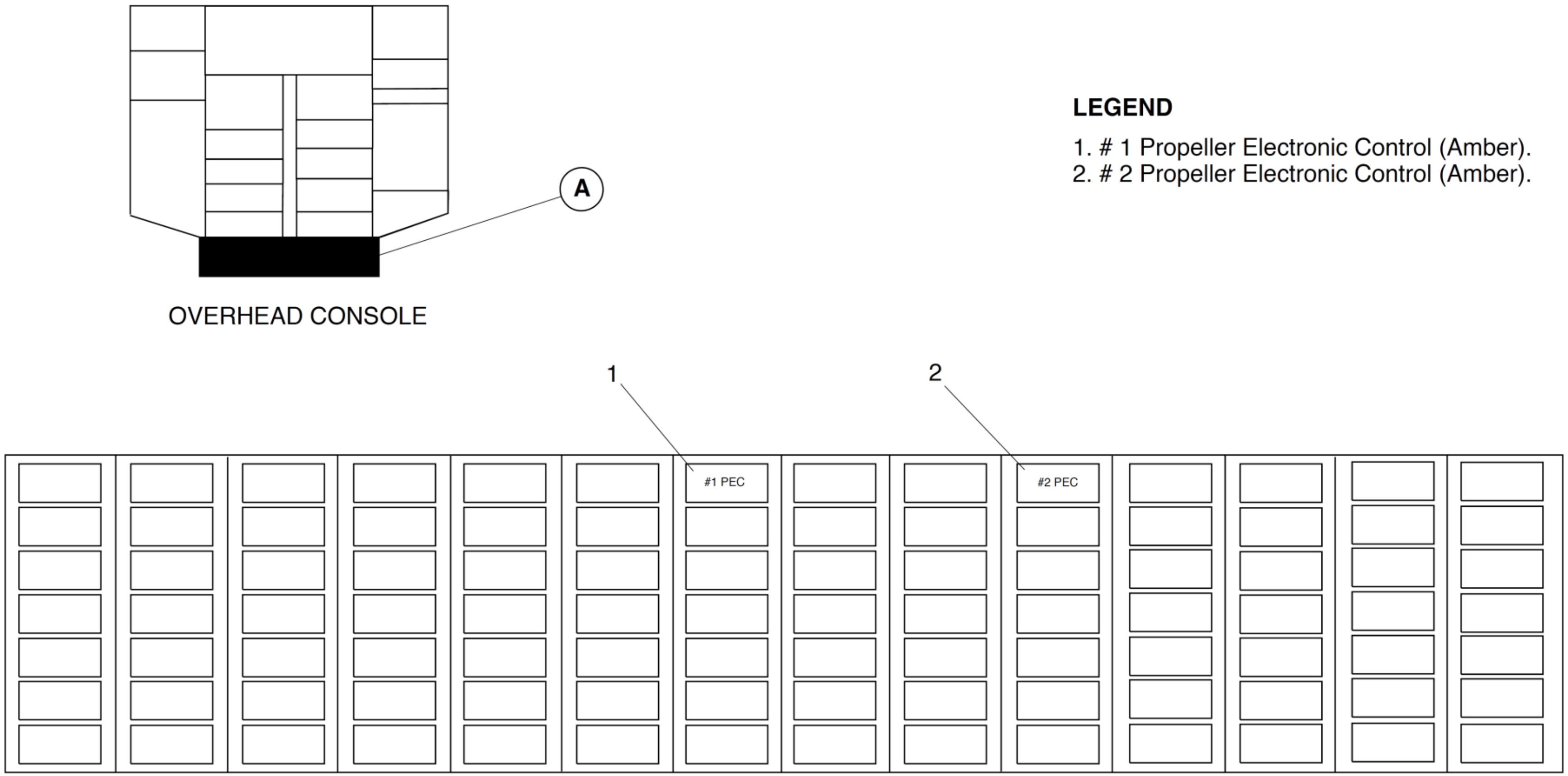
The control lanes get 28V DC electrical power from the left and right essential busses, the active lane also has power from the engine Permanent Magnet Alternator (PMA). The autofeather module is also powered by the essential bus and PMA.

Serial bus links are used to exchange information with the Full Authority Digital Electronic Controller (FADEC). The serial bus links also send maintenance data to the aircraft, through FADEC and the Engine Monitoring Unit (EMU).

The PEC continuously monitors itself and connected equipment for failures. Fault codes are sent to the FADEC and can be shown on the Audio Radio Control and Display Unit (ARCDU) or the Engine Display (ED). Detected faults can be advisory or cautionary. These faults will normally be annunciated on request to the maintenance crew only and are not displayed to the pilots. For faults that are considered non−dispatchable a NO DISPATCH message will be shown on the multi−function display.

Cautionary faults need pilot action. PEC fault discrete will cause the #1 or #2 PEC caution light, or the PROPELLER GROUND RANGE light to come ON during flight.

The PEC is installed in the forward nacelle of each engine, on the spine between the forward and mid frames. Anti−vibration mounts are used at all mounting points, and all connectors point vertically downwards.



LEGEND

1. # 1 Propeller Electronic Control (Amber).
2. # 2 Propeller Electronic Control (Amber).

Figure 61-35 PEC Caution Lights

Propeller Ground-Range Annunciator

Refer Figure 61-36

The Propeller Ground-Range annunciator is on the pilot's glare shield panel. It advises of propeller operation in the low pitch range by the PROPELLER GROUND RANGE lights. There is one for each system.

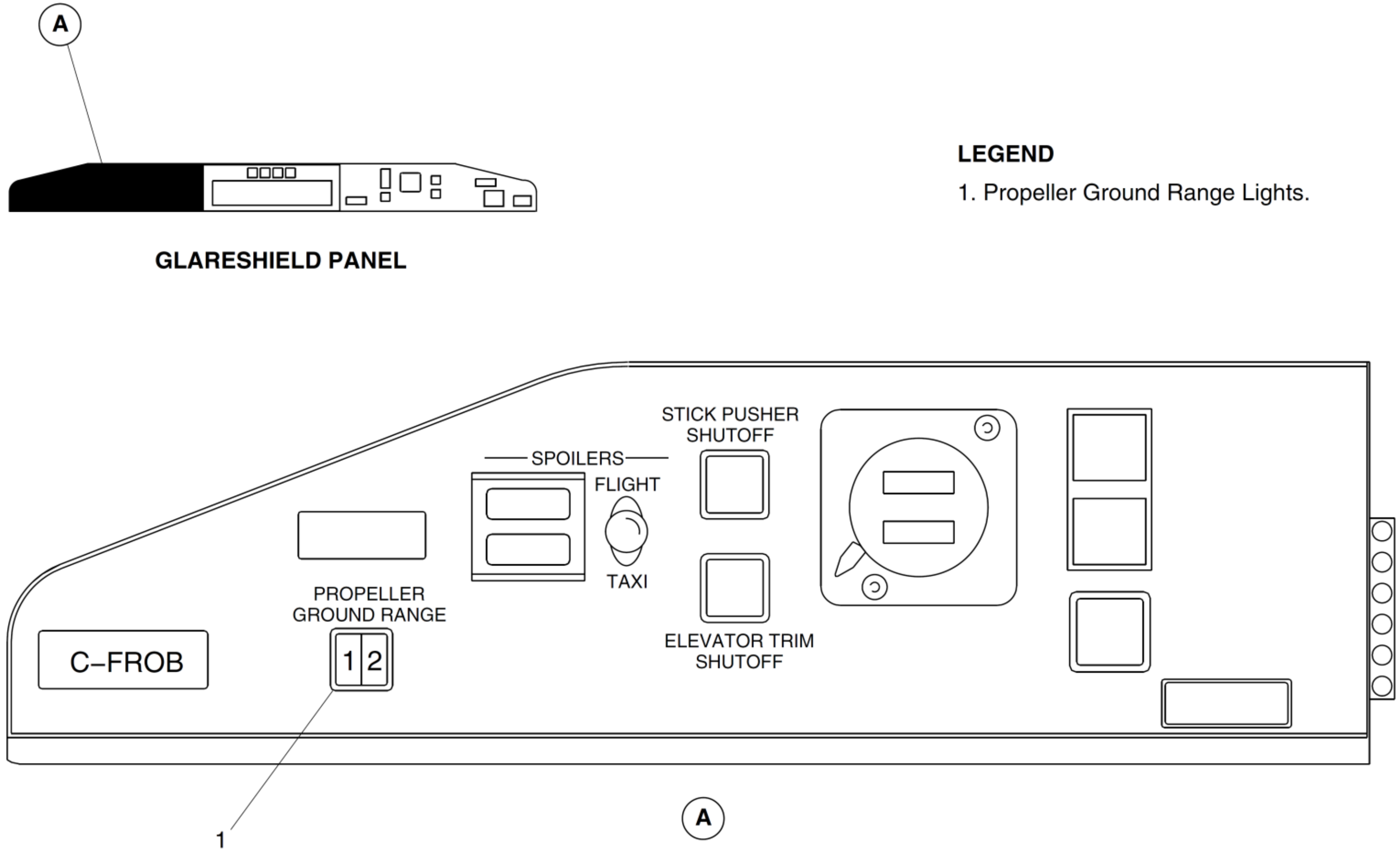


Figure 61-36 Propeller Ground Range Light

Propeller Operation

Ground Start Mode

- Power Lever in Disc position
- Condition Lever in Fuel Off position

Refer Figures 61-37 (Sheet 1 of 2) & *61-38 (Sheet 2 of 2)*

When electrical power is supplied from the essential busses to PEC, the PEC will then supply the excitation current from pins DD and EE to the dual RVDT of the condition lever.

The excitation current enters the RVDT at pins 4 and 5. The RVDT return signal to PEC comes from pins 1, 2 and 3 of the RVDT. It then returns to pins FF, GG and HH of PEC.

With the PLA <33° signal at pin H15 on the 6 – 8 second time delay relay, the relay will energize.

- Power will pass from contact A1 to A2 and from there to pin G of PEC.
- PEC also gets a WOW signal from the PSEU at pin T.
- With these signals PEC will energize the ground beta enable solenoid by sending a discrete output on pin T to pin A on the GBES with a return signal out on pin B to PEC pin J.

Press the starter switch and when N_H is observed move CLA to Start and Feather Position. This will cause:

- Fuel to flow, controlled by FADEC,
- Signal from CLA RVDT to PEC.

PEC will command the servo valve in the pitch control unit using pins N and U of PEC to pins A and B of the servo valve. The servo valve dual torque motor, controlled by the signal from PEC, will position the flapper valve and send high oil pressure, from the propeller pump to one end of the directional control valve. The valve will be positioned to supply coarse pitch oil to the transfer sleeve. From the transfer sleeve the oil will pass through the outer beta tube to the front of the Pitch Change Piston holding the crosshead assembly on the feathering stop.

Unfeather after start

- PLA in Disc position
- CLA to MIN, 900 or MAX.

A signal from CLA RVDT is sent to PEC, and PEC commands the servo valve to supply fine pitch oil to the ground fine oil line from the ground beta enable valve to the transfer sleeve. From the transfer sleeve the oil will pass through the inner beta tube to the rear of the pitch change piston, driving the piston and pitch change shaft forward. This will decrease the propeller pitch to 0° pitch.

As the propeller pitch decreases through 10° the low beta indicator (Prop ground range lights) will come ON. Power is from the Essential. Bus through the light to the advisory lights control unit that provides a ground when it receives a signal from pin A of PEC. From the other side of the pitch change piston, coarse pitch oil will return to the RGB through the outer beta tube, transfer sleeve and the servo valve.

As the propeller pitch decreases towards 0°, Np will increase to 660 N_P where it will be governed by FADEC limiting engine fuel in Np Underspeed Governing mode.

During Taxi with PLA between Flight Idle and disc the propeller will remain in Np underspeed governing.

Increase power for Take Off

- PLA/Rated Power Detent
- CLA/Max

As PLA is advanced above flight Idle towards the rated power detent, power is removed from pin H15 on time delay relay to de-energize the relay. The signal is removed from pin G of PEC and the ground beta enable solenoid is de-energized after a 6 – 8 second delay. Oil is drained from one end of the ground beta enable valve high oil pressure positions it to the inflight position.

PEC gets propeller speed signals from the dual pulse probe assembly input on pins A and B of connectors P/J 20 and 21. PEC also gets reference Np signals from FADEC on ARINC 429 bus. As PLA is advanced Np will increase up to 1020 Np. PEC will now command the servo valve to progressively increase propeller pitch to absorb the increasing engine power and torque will increase to match the torque bug (MTOP or NTOP).

After Take Off

Pilot will retard CLA to 900 position. PEC will command increase propeller pitch. Oil will be directed by the servo valve into the coarse pitch oil line, and through the transfer sleeve to the outer beta tube. The oil from the beta tube goes to the front of the pitch change piston increasing the pitch. The propeller will slow down to 900 Np (MCL)

Constant Speed Mode

With signals from the dual pulse probe assy, and a reference signal from FADEC, PEC can control propeller speed to match CLA input.

PEC will command the servo valve in the PCU to direct oil to the coarse or fine pitch sides of the pitch change piston. Doing this will constantly change propeller pitch to maintain the selected Np.

Operation on the Beta Schedule

The beta feedback transducer sends pitch angle signals to PEC when pitch is near to 27° and below. Above 27° pitch the transducer signal is saturated and therefore not useful.

With PLA >60° the minimum blade pitch allowed by the beta schedule is 27°. The beta schedule is one of PEC's control loops.

- As PLA is retarded < 60° PEC allows < 27° pitch angle as necessary.
- As PLA is retarded to Flight Idle PEC will allow pitch to decrease to 16.5°.
- As PLA is retarded to 33° and below PEC will allow pitch to decrease through disc to maximum reverse -11° to match PLA.

Approach

PEC will control propeller pitch on the beta schedule to match power lever angle.

CLA will be at Max 1020.

Landing and Reverse

When the aircraft lands PEC gets:

- <20 ft AGL from radio altimeters
- PLA <33°.

Note: WOW and AGL signals are in parallel

With these signals PEC will energize the ground beta enable solenoid.

Spring pressure will now move the ground beta enable valve.

This action of the valve will allow fine pitch oil, when commanded by PEC, to enter the ground fine line and the pitch can now be decreased below 16.5° (F.I.)

As the pitch decreases to <10° PEC will command the Propeller Ground Range lights ON.

As the PLA is retarded towards disc and into reverse pitch mode FADEC will increase power and PEC will control propeller pitch to absorb this power and maintain Np between 950 and 1020 Np dependent upon ambient conditions.

The ground beta enable valve when in ground mode isolates the propeller overspeed governor, so maximum Np in reverse is limited by FADEC at 1020 by limiting engine fuel.

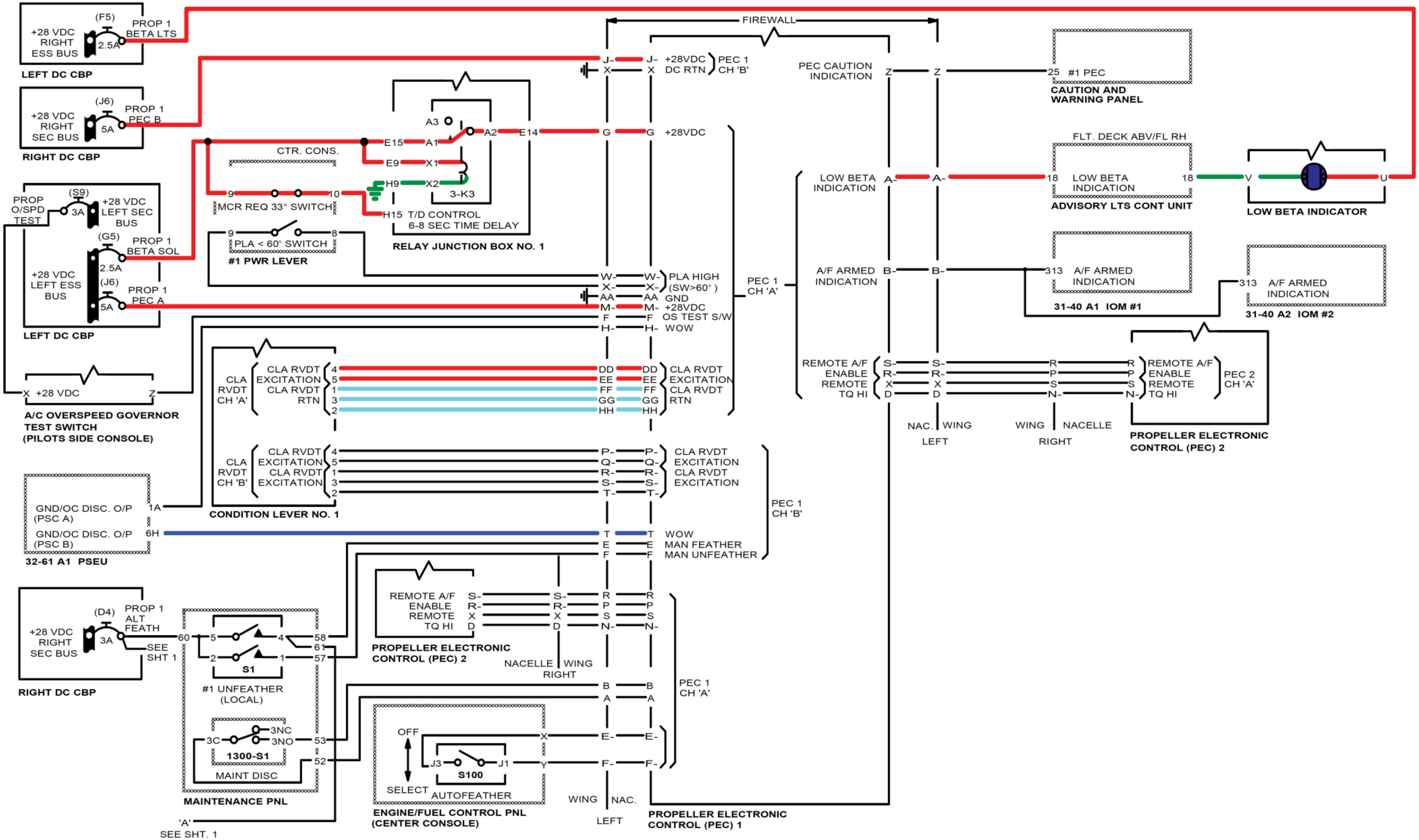


Figure 61-37 Propeller Normal Operation(Sheet 1 of 2)

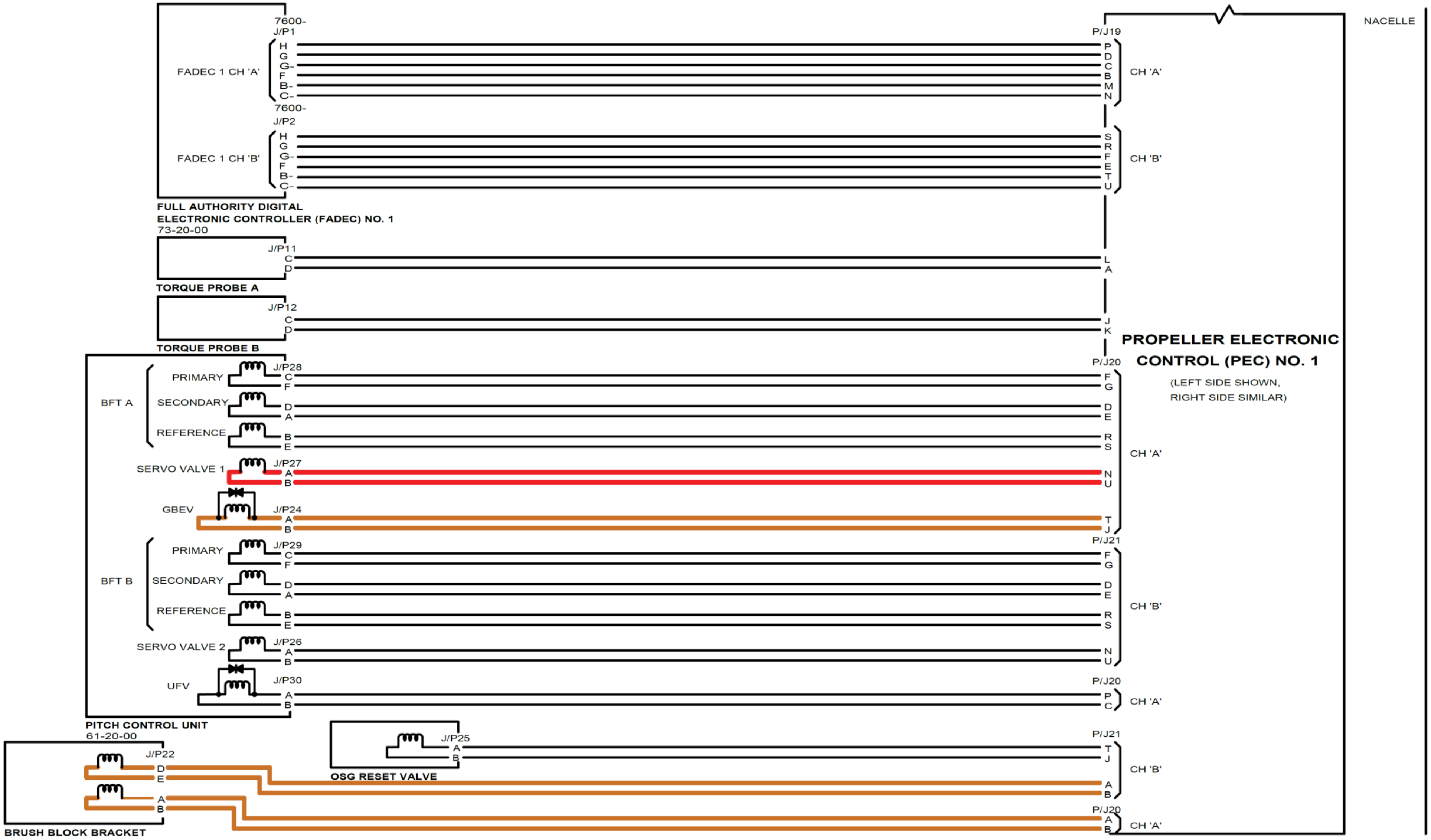


Figure 61-38 Propeller Normal Operation(Sheet 2 of 2)

Autofeather System

Refer Figure 61-39 (Sheet 1 of 2)

Autofeather is selected and armed for Take-Off. The system is deselected by aircrew after Take-Off.

Power is supplied from the secondary bus through diode CR1 to pin H3 on the Autofeather Select Switch.

When the select switch is pushed:

- Power flows from pin H3 to pin H1 and from there to Advisory lights control unit pin 44.
- The unit then outputs power through pin 24 to pins B and C of the autofeather select switch and out through pin G to ground.
- The autofeather SELECT ON switchlight illuminates.

Refer Figure 61-40 (Sheet 2 of 2)

The autofeather system is ARMED when:

- There is power from the Essential bus through PLA >60° switch to pin W of PEC
- PEC receives local TQ Hi signal from FADEC on ARINC bus and remote TQ Hi from the remote PEC on pins X and D
- Autofeather permission from remote PEC on pins S and R
- There are no faults on the autofeather boards in the PECs.

An output from pin B of the PECs to pins 121 and 313 of the IOMs will bring on the A/F ARM message on the ED.

Autofeather Activation

Refer Figures 61-39 (Sheet 1 of 2) & *61-40 (Sheet 2 of 2)*

If PEC now senses a low torque signal from BOTH Npt/Q sensors on the failing engine for a period of 3 seconds it will output power from pin E.

This power will go to:

- pin 58 of the Maintenance Panel
- from pin 58 the power will be output on pin 61
- this power will now go to Time Delay Relay 3-K1 (or2-K3)
- the power will enter on pin A6 and connect to A9
- from pin A9 power will go from contact A2 to A3
- from contact A3 through pin A12 to coil X1 to X2 of relay K3 Feather Pump Contactor.

This will energize the Alternate Feathering Pump.

- Also power from the Sec bus will, through contacts 11 to 12 of relay K3 go to pin 11 of P/J1 of Advisory Light Control
- Power would then be output from pin 11 of P/J2 to pins D and A of switch S101 (or S102) to bring ON the green bar)
- The Alternate Feathering Pump using oil from the reserve oil tank in the RGB will send oil pressure to the Unfeather valve.
- The de-energized Unfeather valve will direct the oil pressure to the top of the Feather Valve.
- This will move the feather valve against the action of its spring.
- This action will open a port into and out of the feather valve through a restrictor R2
- The restrictor will control the rate of feather.
- The oil will pass through the feather valve to the coarse pitch oil chamber of the transfer sleeve.
- From the transfer sleeve the oil will flow into the inner beta tube to the front of the pitch change piston.
- The pitch change piston will drive the blades to feather
- After 30 to 35 seconds the time delay relay will energize:
- Contacts A2 to A3 will open.
- Power will be removed from K3 relay and the pump will stop operating
- Green bar in alternate feather switch will go OFF, power removed from pin D.

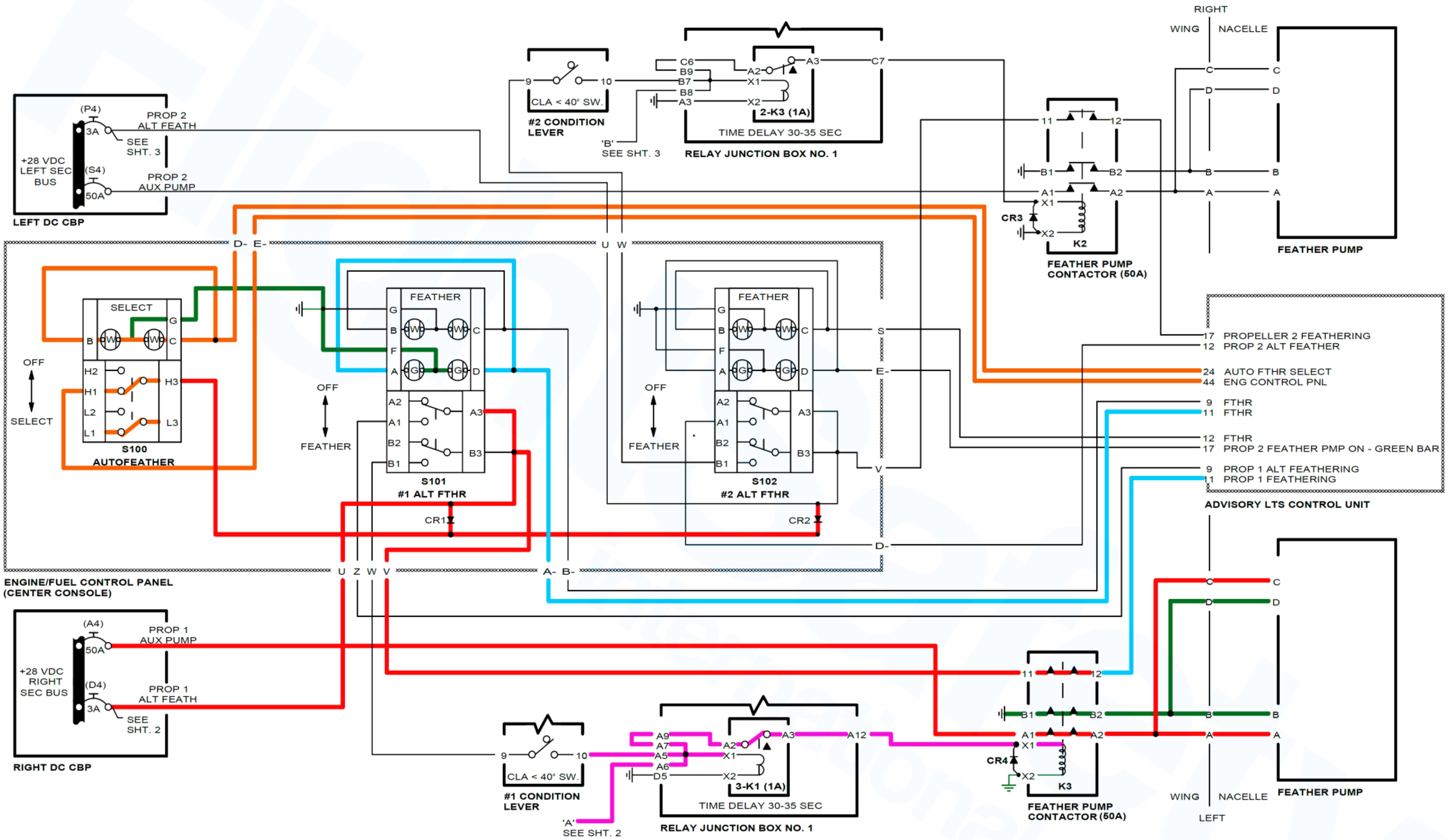


Figure 61-39 Autofeather (Sheet 1 of 2)

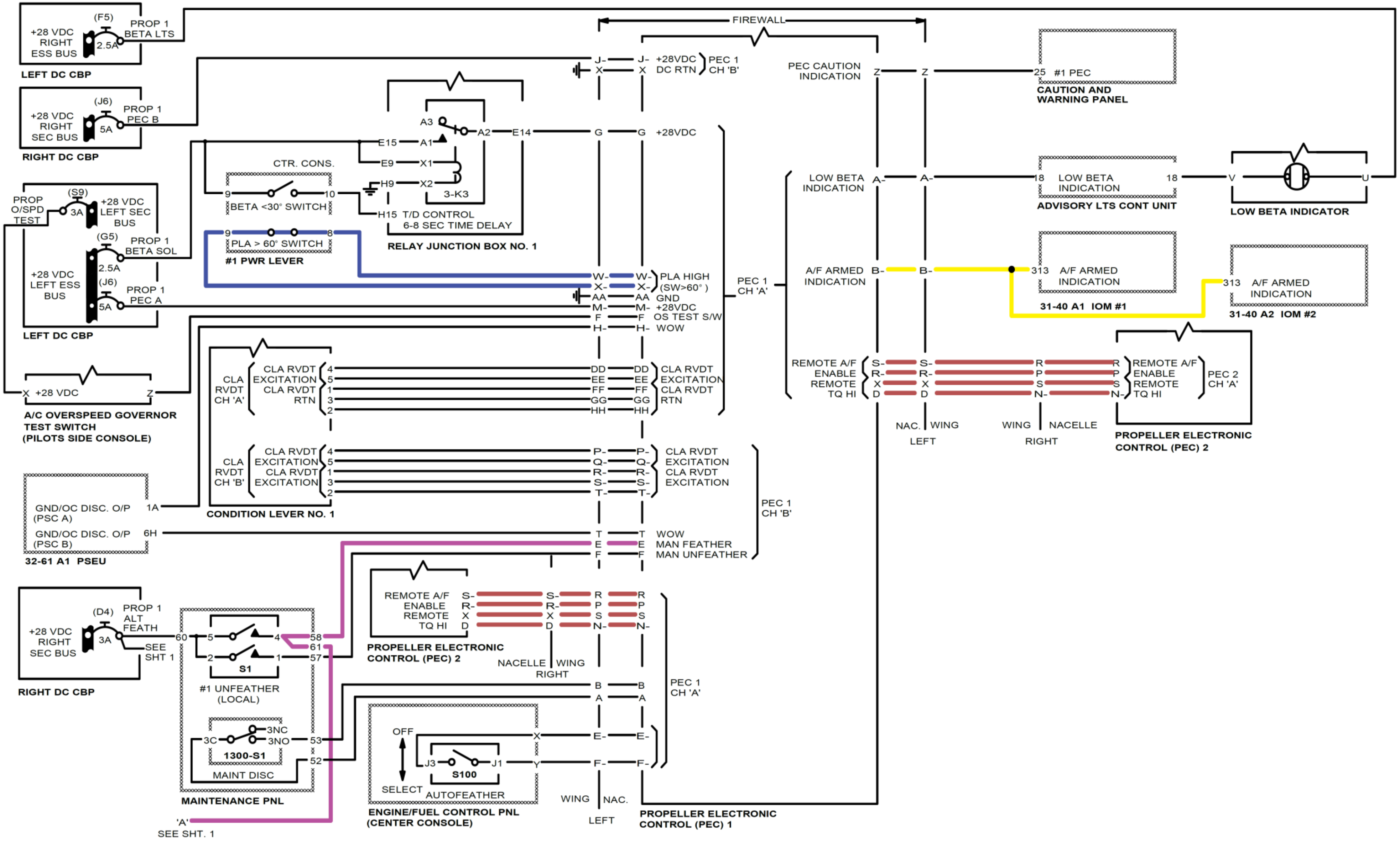


Figure 61-40 Autofeather (Sheet 2 of 2)

Alternate Feather

Refer Figure 61-41

Power from the Secondary bus is applied to pin A3 and B3 on the ALTERNATE FEATHER switch on the PROPELLER CONTROL panel.

When Alternate Feather S101 selected power goes:

- from pin A3 to A1 and
- from there to pin 9 of P/J1 to pin 9 of P/J2, to bring ON the FTHR (white) light on switch S101 through pins B and C of the switch

Power also goes:

- From pin B3 to pin B1 in alternate feather switch
- From pin B1 to pin 9 in CLA < 40° switch
- From 40° switch pin 10 to Time Delay Relay pin A5
- From pin A5 to contacts A2 and A3 of time delay relay
- Output from A3 energizes feather pump contactor K3 closing contacts A1 to A2 and B2 to B1
- This energizes the alternate feathering pump
- Contacts B2 to B1 provide the ground.
- Power also from pin B3 of switch S101 goes to contact 11 on Feather Pump contactor
- From contact 11 to contact12 to pin 11 of P/J 1 advisory lights control
- Output from pin 11 of P/J2 goes to pins A and D on switch S101 to bring ON the green bar (pump operating)

The pump using oil from the reserve oil tank in the RGB will send oil pressure to the Unfeather valve.

- The de-energized unfeather valve will direct the oil pressure to the top of the Feather Valve.
- This will move the feather valve against the action of its spring.
- This action will open a port into and out of the feather valve through a restrictor R2
- The restrictor will control the rate of feather.
- The oil will pass through the feather valve to the coarse pitch oil chamber of the transfer sleeve.
- From the transfer sleeve the oil will flow into the inner beta tube to the front of the pitch change piston.
- The pitch change piston will drive the blades to feather
- After 30 to 35 seconds the time delay relay will energize:
- Contacts A2 to A3 will open.
- Power will be removed from K3 relay and the pump will stop operating
- Green bar in alternate feather switch will go Off, power removed from pin D.

Automatic Underspeed Protection Circuit

Circuit is Armed if:

- NOT Manual feather
- NOT Autofeather
- NOT Alternate Feather
- NOT PLA Ground Beta.

If PEC detects high torque >50% AND low Np< 815 Np from both NPT/Q sensors the AUPC circuit will output a drive fine signal to the servo valve of the Pitch Change Unit.

This drive fine signal will override any other signal coming from PEC and the PEC Caution Light will be ON.

The fine pitch oil, directed by the servo valve, will go to the flight fine pitch port of the Ground Beta Enable valve, and from there to the transfer sleeve.

From the transfer sleeve the oil will flow through the outer beta tube to behind the Pitch Change Piston driving the pitch change mechanism to fine pitch.

The ground beta enable valve will prevent any fine pitch below 16°.

The propeller speed will be limited by the overspeed governor.

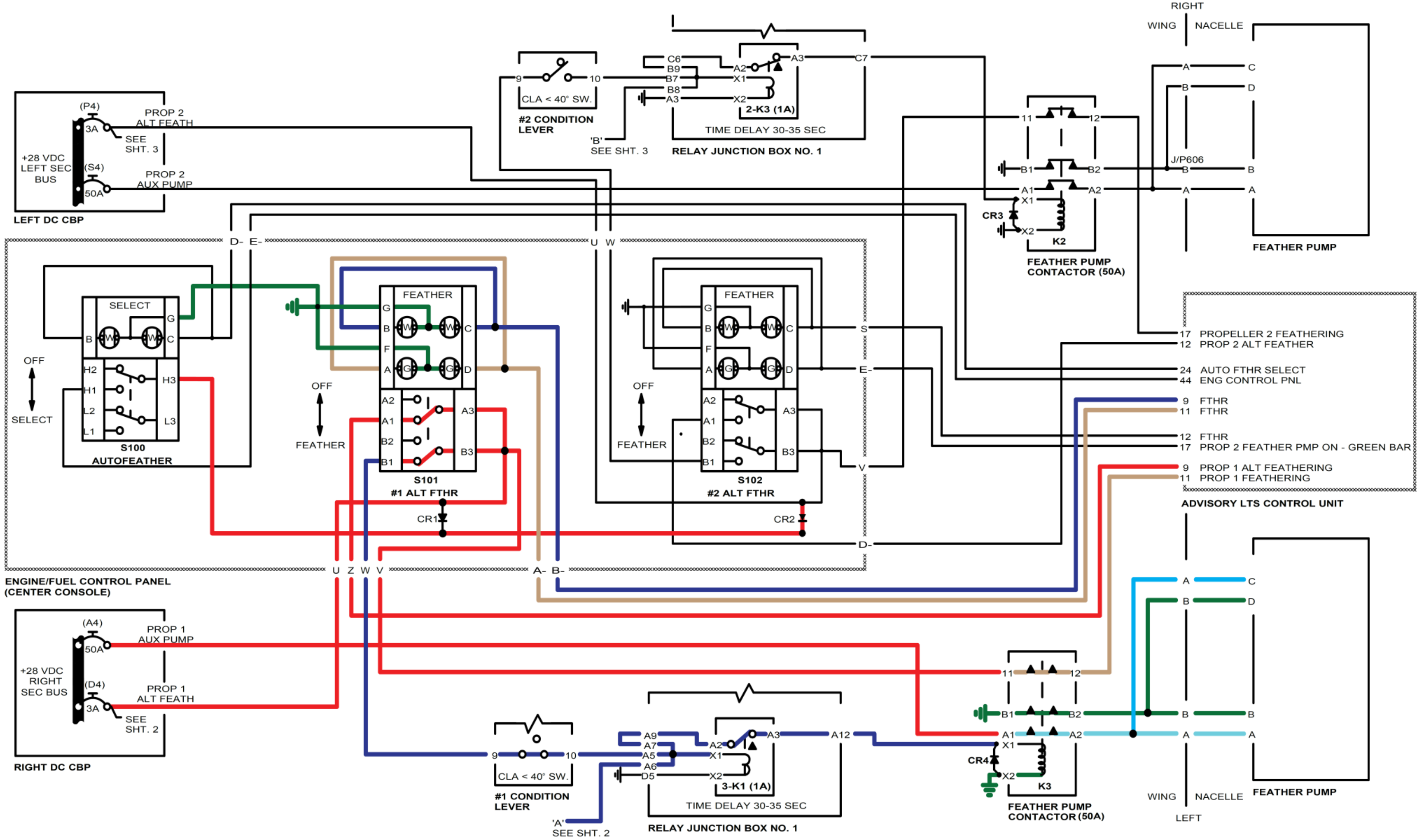


Figure 61-41 Alternate Feather

Overspeed Governor

(Propeller control system NOT in ground beta control)

The flyweights of the overspeed governor are driven by the RGB, and therefore, have a direct relationship to propeller speed.

The action of the flyweights on the governor sliding valve is opposed by governor springs and oil pressure.

When Np is below 1060 the springs and oil pressure are holding the sliding valve in position, to allow oil to flow through the governor to the GBEV.

From the GBEV the oil flows to the second stage of the servo valve where it is then controlled by the servo valve, which is controlled by PEC to control N.

If the propeller overspeeds to 1060 Np the flyweights are now providing enough force to overcome the springs and oil pressure.

- The flyweights move outwards which raises the sliding valve.
- This action cuts of oil to the second stage of the servo valve.
- No oil supply to the propeller pitch change mechanism results in the counterweight assemblies now becoming the controlling force on the blades.
- Blade pitch will be increased to control the propeller at 1060Np
- If the pitch increase caused by the counterweights causes Np to decrease below 1060 the sliding valve will re-open and Np control will revert to normal.

Overspeed Governor Test

Refer Figure 61-42 (Sheet 1 of 2)

Select overspeed governor test switch:

- Power from Sec bus to pin X of the switch (permanently)
- from pin X to pin Z
- From pin Z to pin F of PEC

Refer Figure 61-43 (Sheet 2 of 2)

Output from PEC P/J21 pin T to Overspeed Governor Reset solenoid
Return signal from OSG reset solenoid to PEC pin J of P/ J21.

The energized OSG reset solenoid allows the oil pressure to drain away from the top of the governor.

This means that the flyweights only have to overcome the force of the governor springs.

This action resets the governor down to approximately 860 Np. The action is then as described previously for the overspeed governor but at a lower Np.

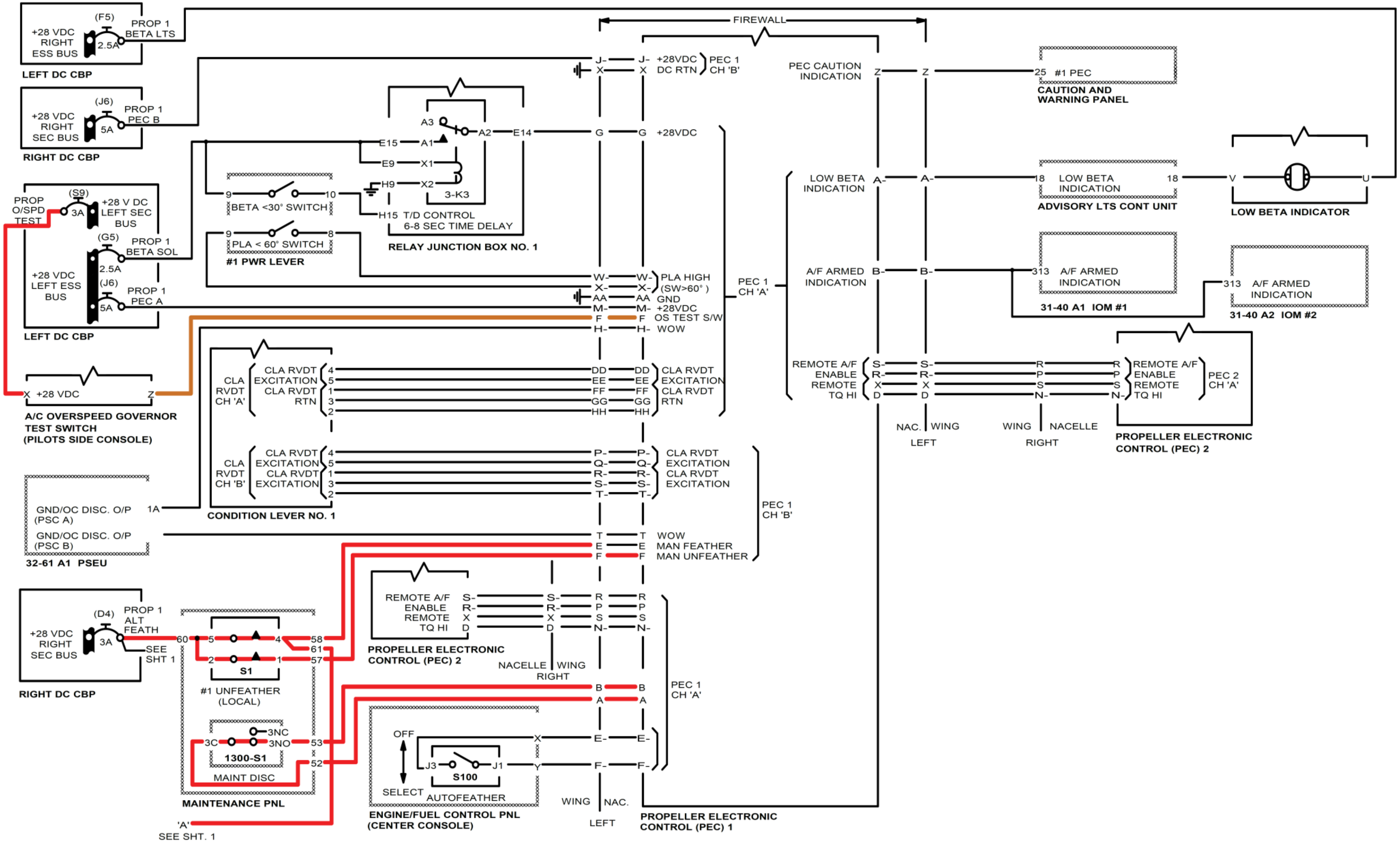


Figure 61-42 Propeller Overspeed Governor Test (Sheet 1 of 2)

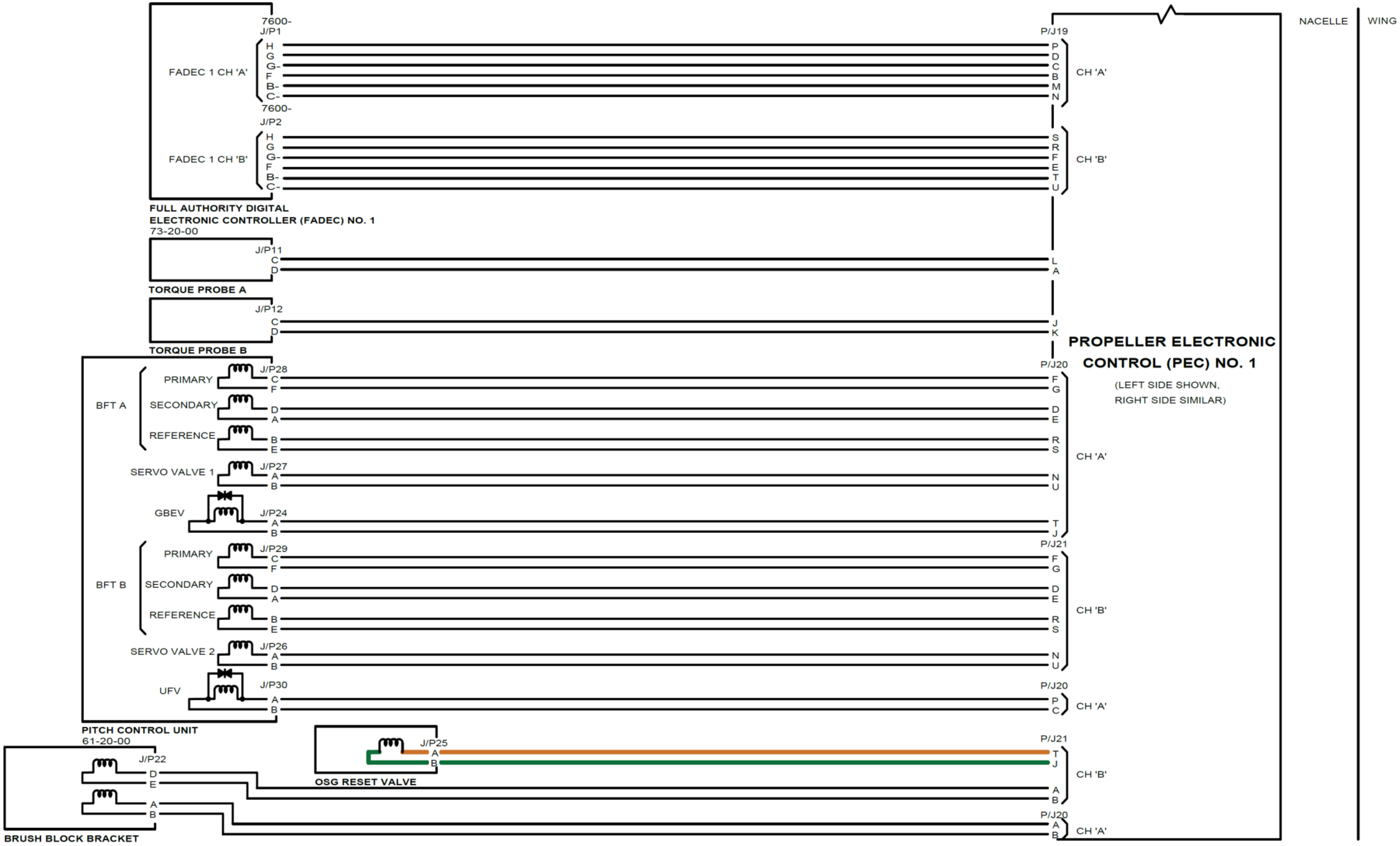


Figure 61-43 Propeller Overspeed Governor Test(Sheet 2 of 2)

Maintenance Unfeather

Refer Figures 61-44 (Sheet 1 of 2) & 61-45 (Sheet 2 of 2)

Prerequisites:

- WOW DETECTED
- No propeller rotation
- Maintenance mode selected.

Power is always supplied from Sec bus to pins 5 and 2 of the Unfeather switch

With all of the above and unfeather switch selected on the Maintenance Panel.

- Power goes from contact 5 to contact 4 of the switch and then to pin E of PEC
- Power also from contact 2 to contact 1 of the switch to pin F of PEC
- With these signals PEC will energize the solenoid of the Unfeather valve
- The power output from contact 4 also goes to pin 61 of the unfeather switch.
- From pin 61 the power goes to the time delay relay pins A5 and A9
- From pin A9 to contacts A2 to A3 to coil X1 and X2 of relay K3
- This will connect power from the Sec bus to the pump. Power from Sec bus also through 11 and 12 of K3 relay to pin 11 advisory lights control.
- Advisory lights control will output power on pin 11 P/J2 to contacts D and A on switch S101 (or S102)
- The green bar in the switch will illuminate because the pump is operating.

The oil output from the alternating feathering pump will go to the Unfeather Valve:

- The solenoid in the unfeather valve has been energized by PEC.
- Oil through the solenoid ball valve to end of unfeather valve.
- Unfeather valve moves against action of its spring
- This connects output port of valve to servo pressure line
- Servo valve commanded by PEC to fine pitch.
- Oil pressure from pump directed by servo valve to fine pitch chamber of transfer sleeve
- From transfer sleeve, through outer beta tube to rear of pitch change piston
- This fine pitch oil can drive the propeller to Maximum Reverse.

When the Time Delay Relay times out, 30 – 35 seconds, the relay will energize opening contacts A2 to A3:

- This will remove power from coil X1 and X2 in K3 relay
- Relay contacts A1 to A2 will open
- Alternate feathering pump will stop
- Contacts 11 and 12 will open in K3 relay
- Advisory lights Control will remove power from green bar in switch S101 (or S102).

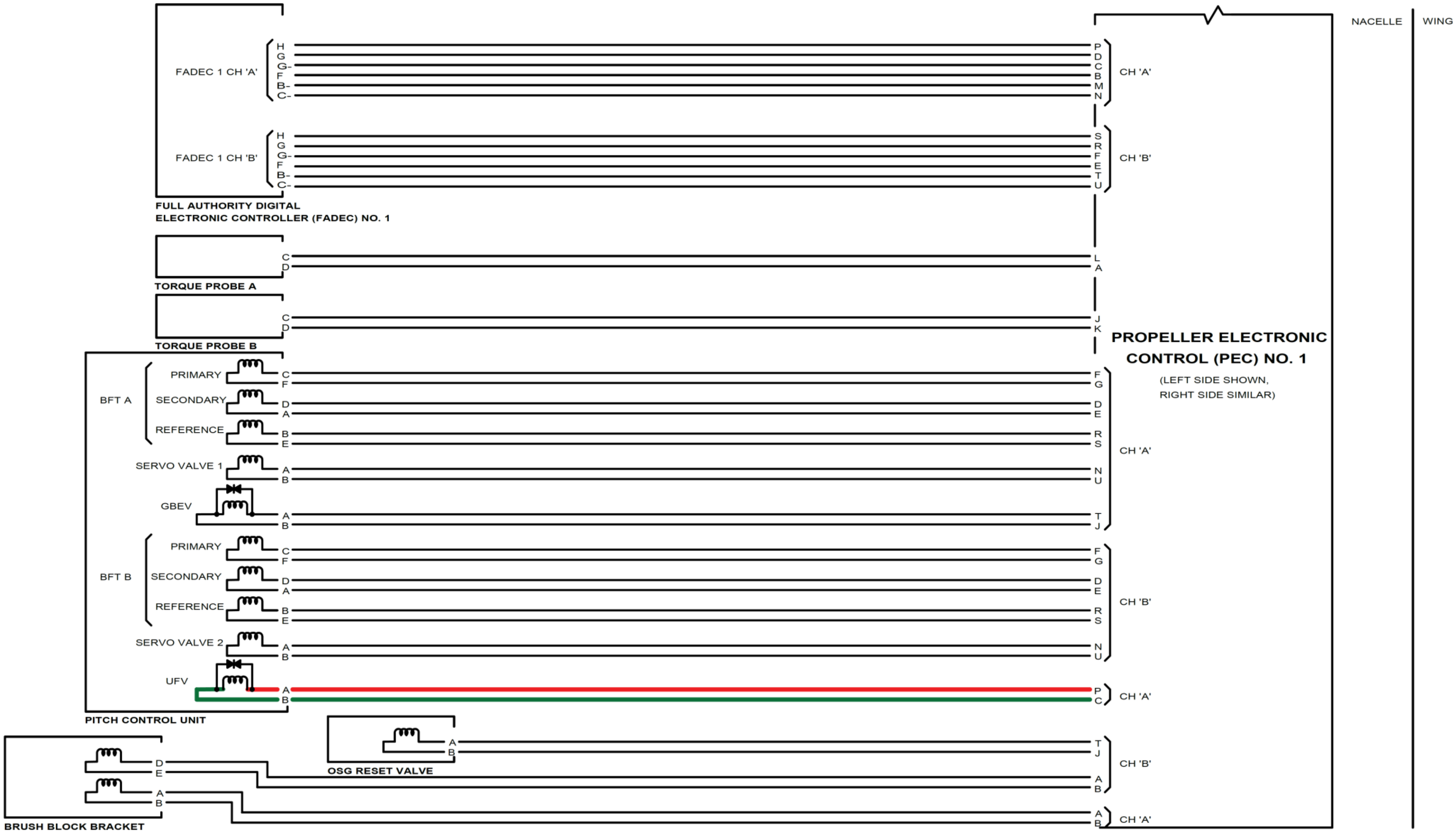


Figure 61-44 Maintenance Unfeather (Sheet 1 of 2)

Figure 61-45 Maintenance Unfeather (Sheet 2 of 2)